

GE Healthcare

Signa Rear Pedestal Stabilization Installation

Technical Publication



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Rev 3

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WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・このサービスマニュアルには英語版しかありません。
- ・GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
- ・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

Revision History

Date	Rev	Change Description
17-Jul-2007	1	Initial Release
29-May-2009	2	Section 4-5-3: Added installation of two washers on each brace (PQR 13225736).
10-Sep-2010	3	Sections 1 & 2: Revised Ferrous Material Handling warning. Added Notice for MR safety training. Section 4: Added Ferrous Material Handling warnings for removing Rear Pedestal from magnet room. (CAPA 1859293)

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SECTION 1 – OVERVIEW

1-1 INTRODUCTION

This manual provides instructions to install the components to connect the Rear Pedestal to the Magnet Enclosure.

This Direction is included with 5198961, Rear Pedestal Assembly, and is included with the following catalogs:

M3335BC – Early LCC WideOpen IIQ Upgrade

M3335BD – LCC WideOpen IIQ Upgrade

This upgrade includes braces and hardware to connect Rear Pedestal to Magnet



FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

NOTICE!

During this hardware upgrade, the magnet may be ramped to full field strength. Individuals working in the magnet room must be MR safety trained.

1-2 SITE READY CHECK

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion.

1-3 PRODUCT DELIVERY INSTRUCTIONS AND PACKING

The “Shipping Document” lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short. Labels that are attached to the outside of packing boxes summarize box contents. PDI (Product Delivery Instructions) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number. Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of the box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with “Technical Publication” boxes for a list of delivered documents.

1-4 UPGRADE SCHEDULE CONSIDERATIONS

FMI, periodic maintenance, installation of prerequisite upgrades and/or options will directly add to the man-hour and system down-time total. Be sure that all planned activity is considered when predicting the return of system to customer. Also verify Application Specialist is scheduled at the end of the installation to instruct users. Additional time must also be scheduled to instruct users following completion of upgrade.

1-5 DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period. File a report with:

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

1-6 DISCARDED MATERIAL RETURN POLICY

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1-6-1 MATERIAL POLICY OVERVIEW

1-6-1-1 Returned Material

Per GE Healthcare Policy 3.9: all requested material is to be returned, as directed, intact and promptly either to GoldSeal or Renewable Resources. Do not remove or swap any parts from the material to be returned.

1-6-1-2 Disposed Material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

1-6-2 MATERIAL TO BE RETURNED

Return all parts, cables, modules, and equipment de-installed for the HDx upgrade to GoldSeal.

1-6-3 RETURN KIT

Packaging and/or transportation details can be obtained through the local Installation Specialist. If unavailable, call GoldSeal. The following material may have been provided to facilitate return of material.

- Shipping labels
- Waybill
- Address labels
- Reuse appropriate containers that new components were shipped to site for return of above material.
- Packing material

1-6-4 DEINSTALLATION PROCEDURES

Refer to instructions in this manual for any deinstallation procedures of the site.

1-6-5 REGION SPECIFIC RETURN INFORMATION

1-6-5-1 Sites in Americas

A GoldSeal return pre-printed BAX waybill and mailing label(s) may be included with the upgrade kit. The upgrade returns kit (p/n 2352804) consists of:

- 2358604 – Waybill
- 2357240 – GoldSeal Return Labels
- 2357241– Upgrade Return Envelope (waybill and return labels)

Gold Seal Address and contact for coordination of transportation arrangements and packaging:

GE Medical Systems – GoldSeal
3114 North Grandview Blvd
Waukesha, WI 53188
Phone: 1-877-251-4468 or 262-513-4147

For non-GoldSeal Returns, utilize GEMS Renewable Resources. For coordination of transportation arrangements and packaging:

Phone: (414)-747-6997 or Dial Comm 8*579-6997
Fax: (414)-747-6855
Field ASPEN: 1-800-525-1516, Box #26041

1-6-5-2 Sites in Asia

At this time, system components are not returned for systems in Asia, but whole systems must be returned from complete system (forklift) upgrades.

Return procedure for GEMS-Asia countries:

Please contact your Regional Service Headquarters to get instructions.

1-6-5-3 Sites in Europe

The GoldSeal Returns Coordinator contacts each site individually and reviews the material to be returned and the logistics for the return. A list of material and a printable shipping label is sent to the site's primary FE, service providers, and forwarders.

GoldSeal Europe return information:

GEMS-E – GOLD SEAL OPERATION & RECYCLING CENTER
GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL
Quai 45 à 49 – 91090 LISSES – FRANC
Tel: (+33) 1-60-91-57-35 or (+33) 1 64 97 55 41
Fax: (+33) 1-60-86-12-31

For non-Gold Seal returns in Europe:

European Renewable Resources Center
GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL2
Quai 45 à 49 – 91090 LISSES – FRANCE
TEL = (+33)-1-64-97-55-41 FAX = (+33)-1-60-86-12-31

1-6-6 PRODUCT LOCATOR INFORMATION

Process product locator information for all serialized components removed from the system according to policy. Refer to Section 1-7.

1-7 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at: <http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: [Product Locator Support Central Page](http://supportcentral.ge.com/15563) – <http://supportcentral.ge.com/15563>

2. One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1-8 UPGRADE SEQUENCE

Follow the system Upgrade Flow chart in SECTION 2 for best installation efficiency.

Note

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

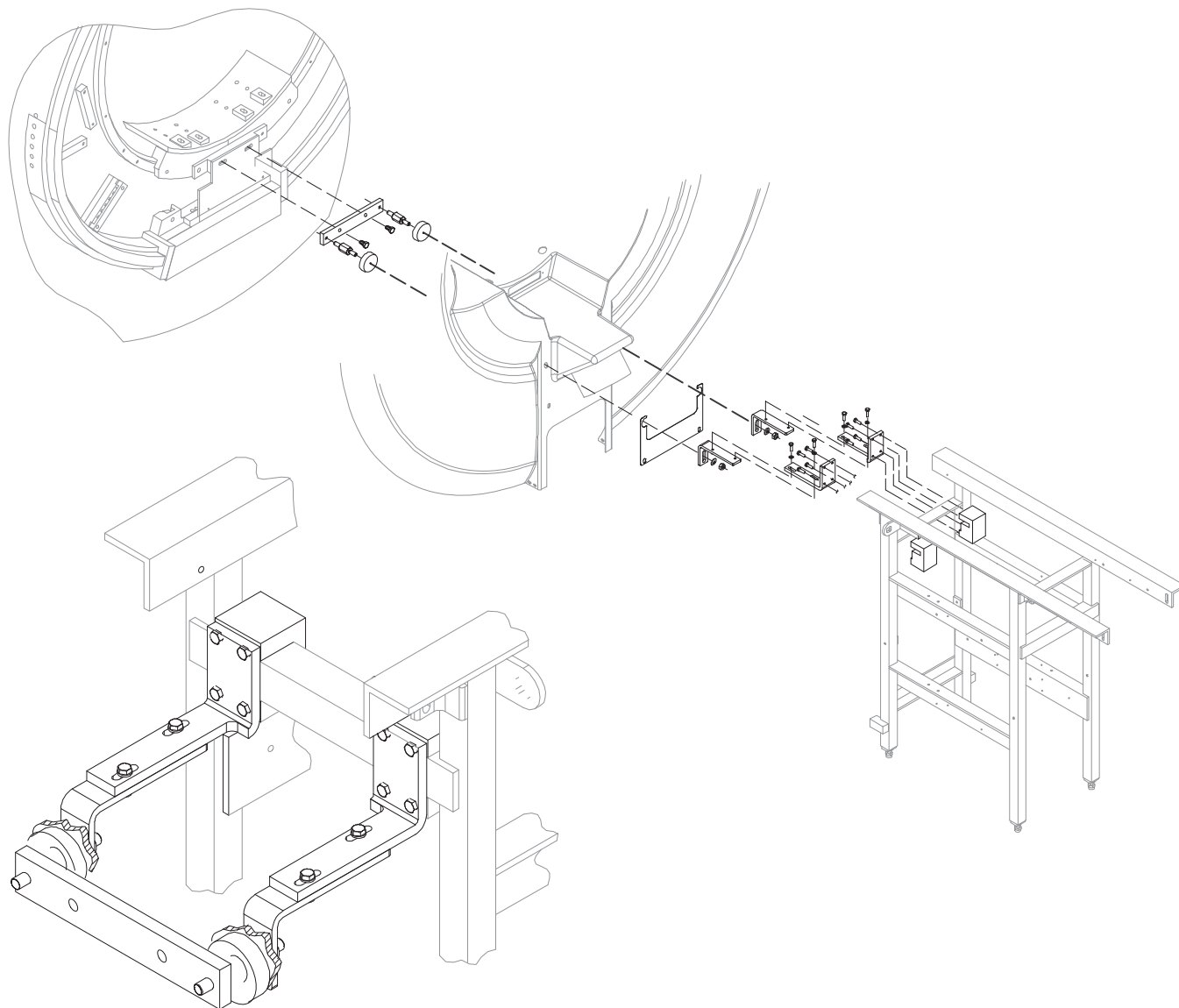
Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

The general sequence of events is:

- Verify that all required upgrades have arrived complete and site is ready to shut down.
- Remove any obsolete equipment and cables.
- Install new equipment and modify existing components.
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Perform Functional Checks, System calibrations, and System performance checks
- Replace covers, complete final tasks, and return site to customer.

1-9 UPGRADE SYSTEM COMPONENTS

Illustration 1-1 shows the major equipment modified for this upgrade.



REAR PEDESTAL BRACE ASSEMBLY
ILLUSTRATION 1-1

SECTION 2 – UPGRADE INSTALLATION PROCESS

2-1 SITE READINESS

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to upgrade the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

**WARNING!**

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS UPGRADE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

2-2 UPGRADE PROCEDURE

The following upgrade flow charts contain information for installing the Rear Pedestal Stabilization Kit on a system with a Split Bridge or older style Long Bridge.

The upgrade flow chart should be followed for an orderly and efficient installation. Note that procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

This upgrade flowchart has been developed assuming that all Signa HDx equipment has been delivered. **Make sure that every part required is available before starting.**

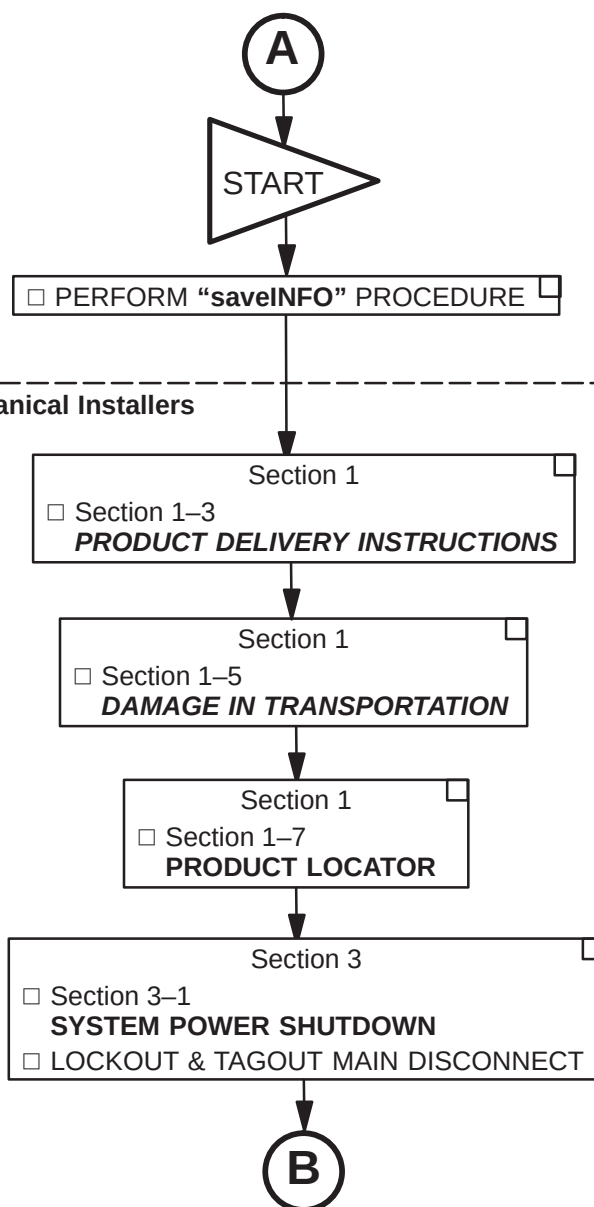
2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART

Procedures to be Completed by Field Engineer



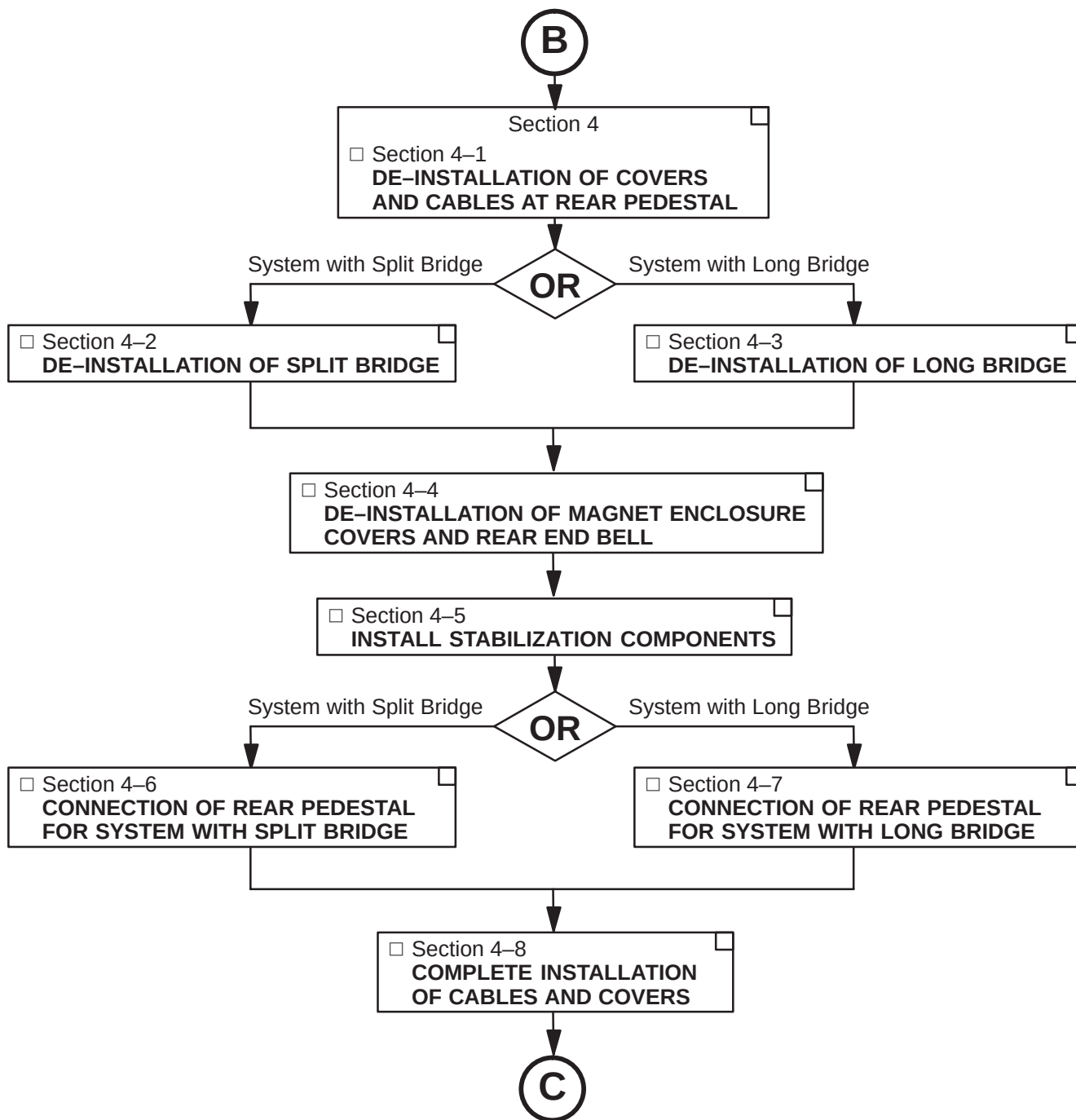
IMPORTANT : PER ISO
9001 REQUIREMENTS

Procedures to be Completed by FE and/or Mechanical Installers



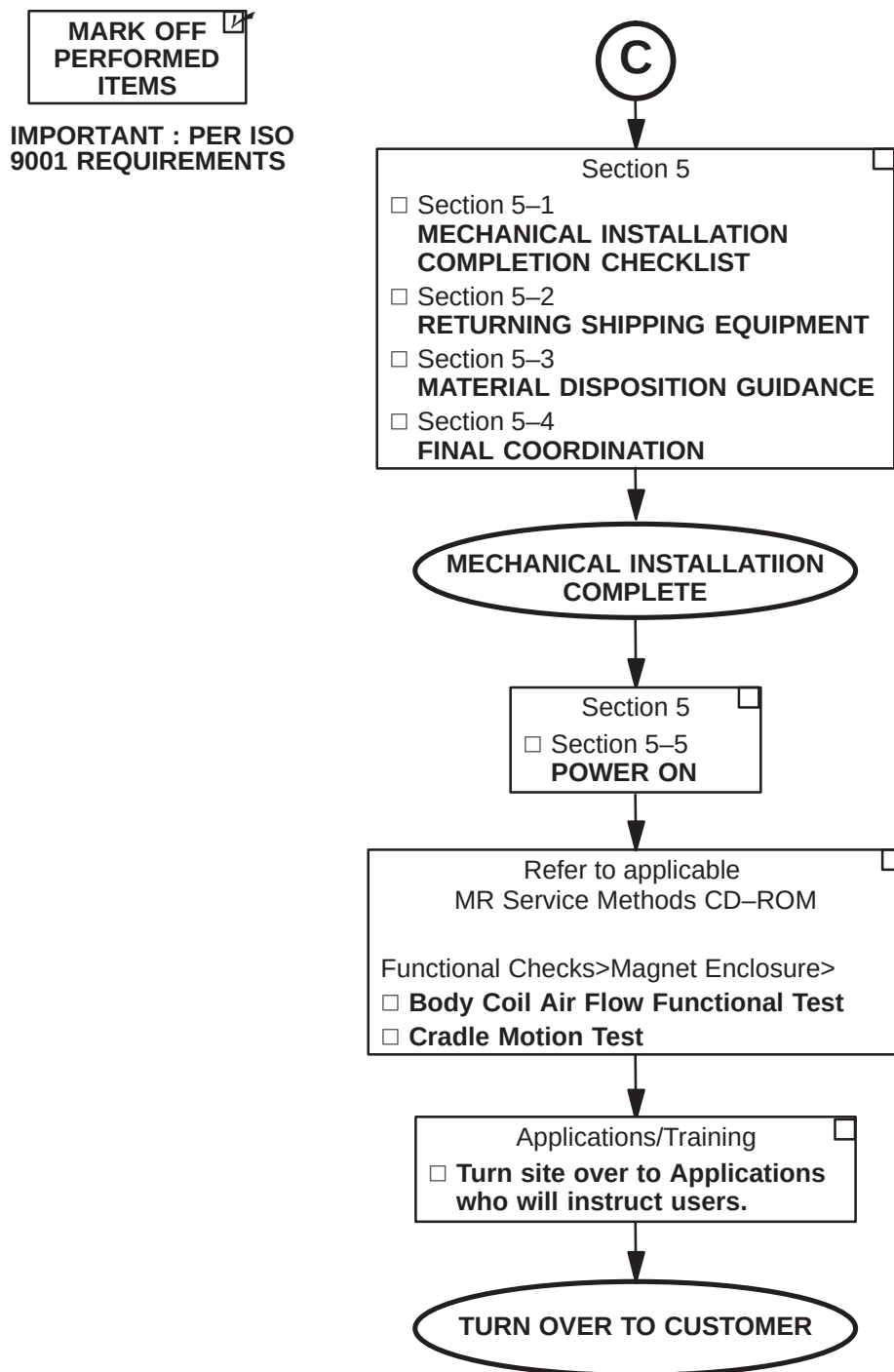
2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Procedures to be Completed by FE and/or Mechanical Installers



2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Procedures to be Completed by FE and Applications



SECTION 3 – POWER OFF

3-1 SYSTEM POWER SHUTDOWN



FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1. Make sure that site customer has archived all patient information. Execute "Save Info".
2. Verify that System performance baseline measurements have been obtained, especially:
 - SPT Full Test Mode
 - White Pixel Test
 - LV Shim
 - IIS
3. Record center frequency.
4. Ensure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
5. Press the POWER OFF button on the PDU front control panel.
6. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
7. Locate and shut off main disconnect supplying power to the PDU.
8. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
9. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring

SECTION 4 – UPGRADE INSTALLATION

This section contains instructions on installing the Rear Pedestal Stabilization components for sites with a Split bridge and for sites with the older style Long Bridge. Refer to installation flowchart in Section 2 for guidance on which procedures to complete for the site being upgraded.

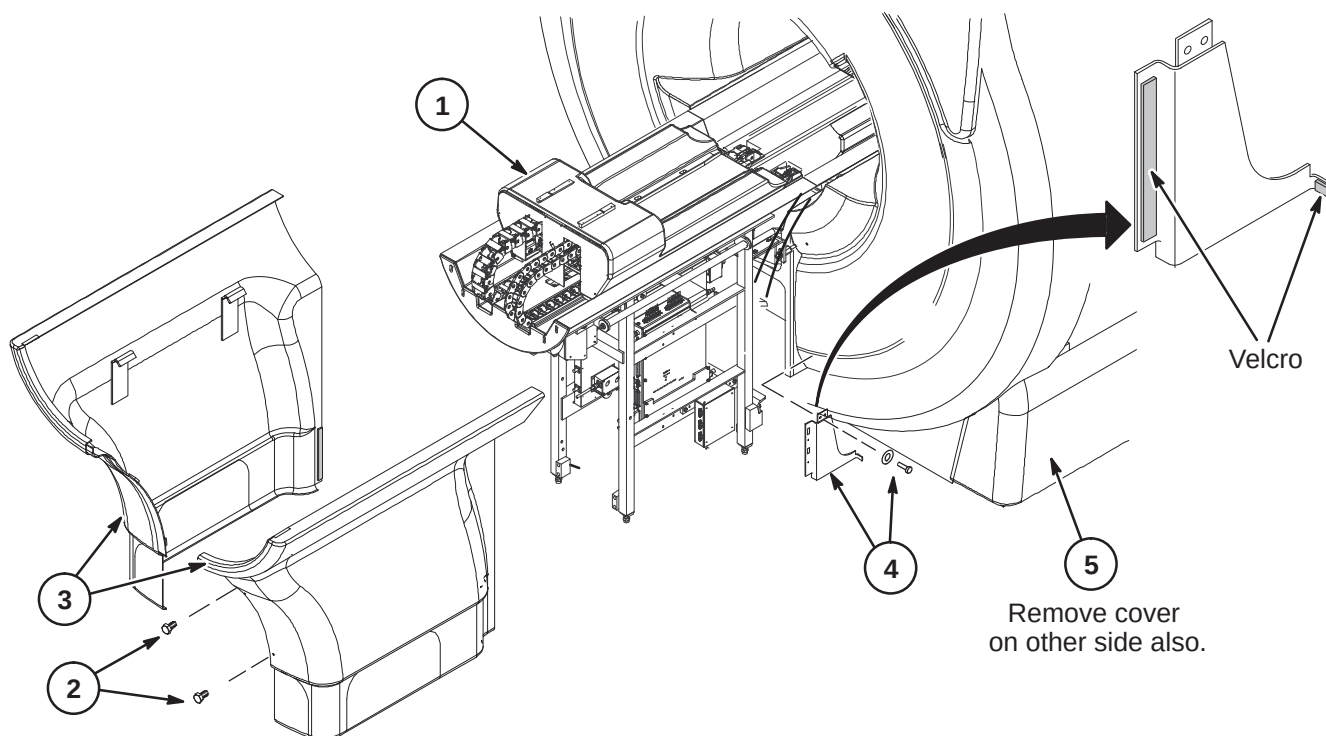
4-1 DE-INSTALLATION OF COVERS AND CABLES AT REAR PEDESTAL

The illustrations in Section 4-1 show a system with a dual track LPCA and split bridge. The de-installation procedure for a system with a single cable track and split bridge or the older style one-piece long bridge would be the same.

4-1-1 De-Install Rear Pedestal Covers

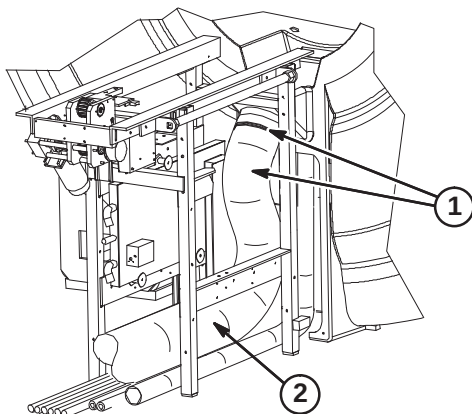
All hardware and fasteners removed in this procedure will be re-installed.

1. Move LPCA to rear of Bridge.
2. Unfasten left and right covers by removing two M6 x 10mm Pan Head screws as shown below.
3. Lift and remove left and right Rear Pedestal Covers from horizontal support bars. Covers are attached to base trim covers with Velcro.
4. Remove two M4 x 8mm screws and washers from Rear End Bell covers on each side of Rear Pedestal and remove covers. Covers are attached to base trim with Velcro.
5. Remove left and right rear lower trim covers.

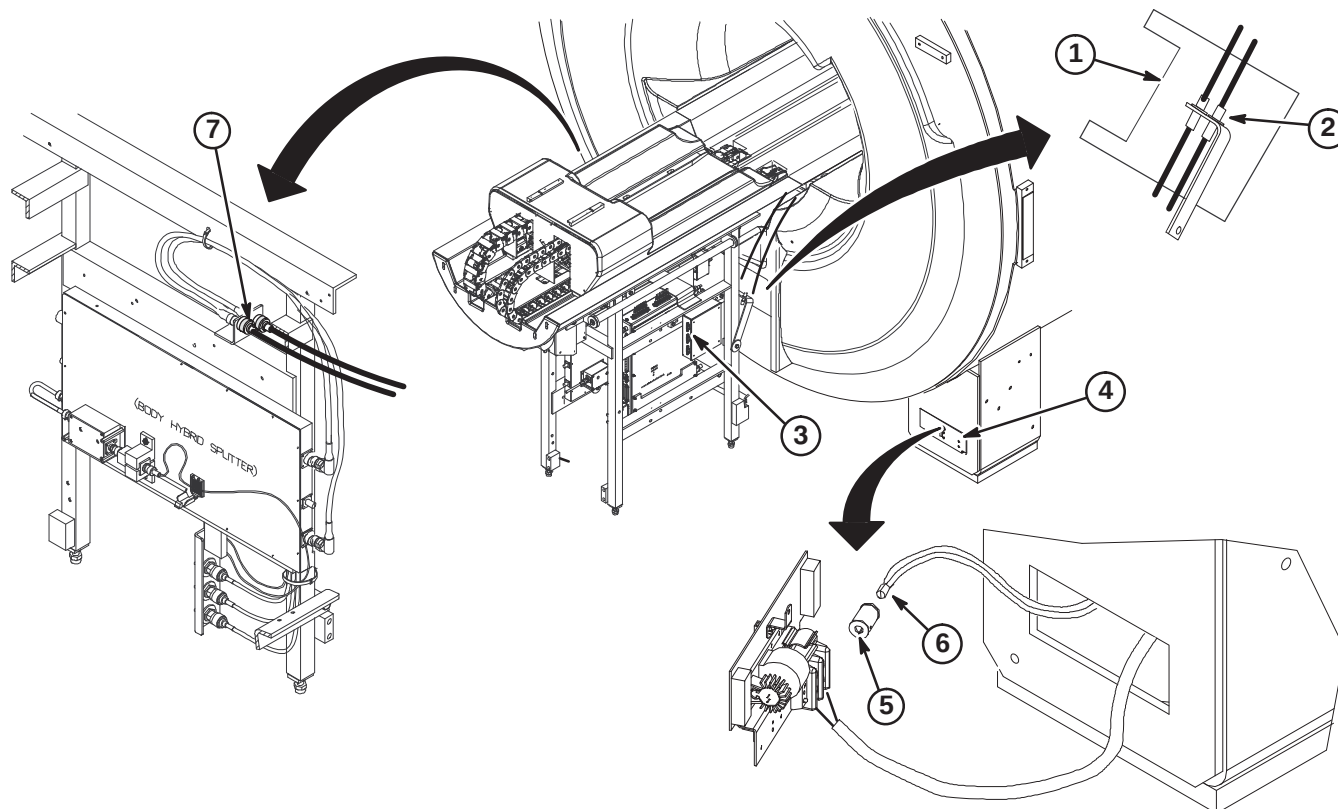


4-1-2 Disconnect Air Hose

1. Loosen 6 inch clamp and carefully pull 6 inch hose from Rear End Bell housing. Clamp will be reused.
2. Pull 6 inch hose back thru Rear Pedestal.

**4-1-3 Disconnect Cables to Rear Pedestal Area**

1. Remove and save Connector Wrap from Bias cable connections.
2. Disconnect RF body coil Bias cables from J5 and J6 on bracket.
3. Disconnect cables from J8 and J9 on Patient Communication Box.
4. Carefully pull out Bore Light Assembly that is secured in place with Velcro in rear leg of magnet.
5. Pull out Bore Light fiber optic bushing from clip on Bore Light assembly
6. Carefully pull out fiber optic bundle from bushing.
7. Disconnect RF body coil I and Q cables from bracket.



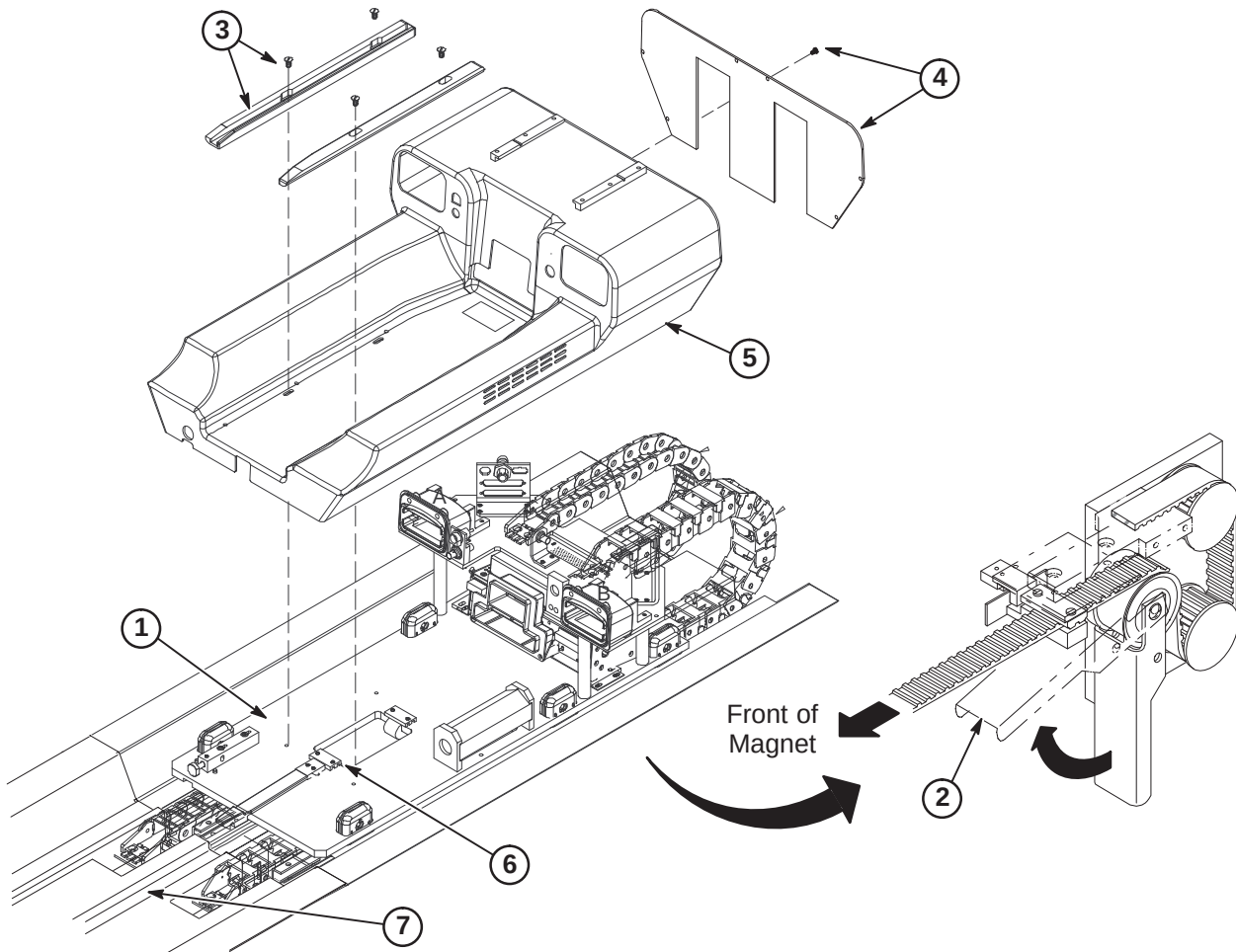
4-2 DE-INSTALLATION OF SPLIT BRIDGE

The procedures in this section provide instructions on removing the Split Bridge. For those systems that have a Long Bridge that needs to be removed, refer to Section 4-3.

4-2-1 Remove LPCA Cover

The dual-track LPCA is shown below. The procedure would be the same for a single-track LPCA.

1. Make sure LPCA has been moved to rear of Bridge.
2. Release tension arm.
3. Remove the four screws that attach the cover to the base plate. Save screws and slides for re-installation at end of upgrade.
4. Unfasten and save six screws and Rear Cover.
5. Carefully remove cover from base plate assembly.
6. Unfasten only one clamp and disconnect end of drive belt that is routed to front of magnet.
7. Carefully pull drive belt from front of magnet in preparation of removal of Front Split Bridge.

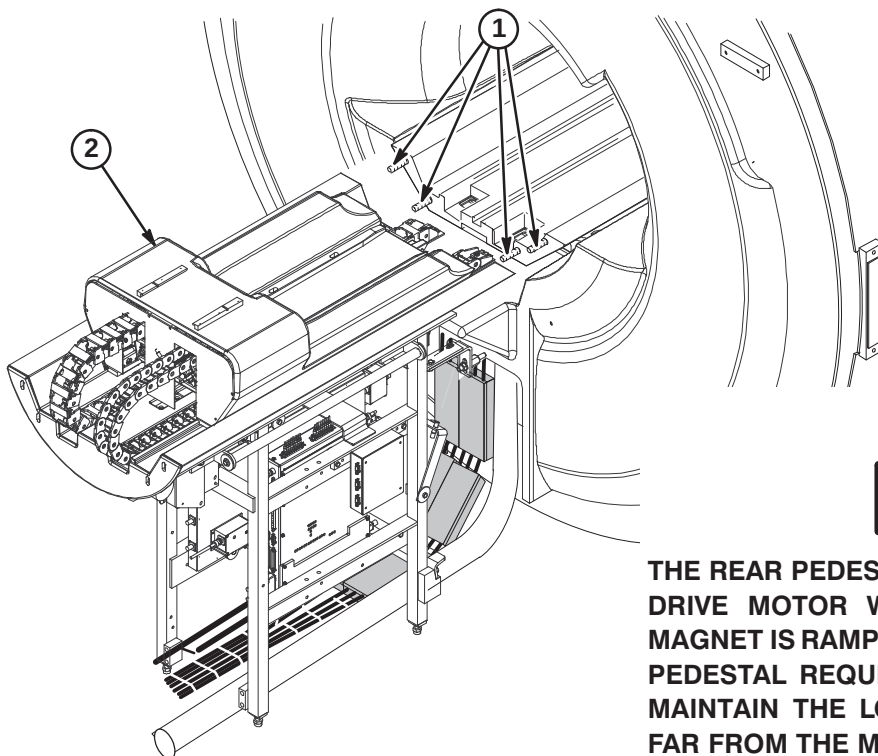


4-2-2 Unfasten and Move Rear Pedestal and Rear Split Bridge



The weight of the Rear Pedestal requires at least THREE people to move Rear Pedestal from the rear of the magnet.

1. From underneath Bridge, unfasten, remove, and save the four hex nuts and washers that attach the front and rear split bridge.
2. Using at least three people, move the Rear Pedestal back, away from the magnet enclosure so that there will be enough room to unfasten and remove the Rear End Bell.

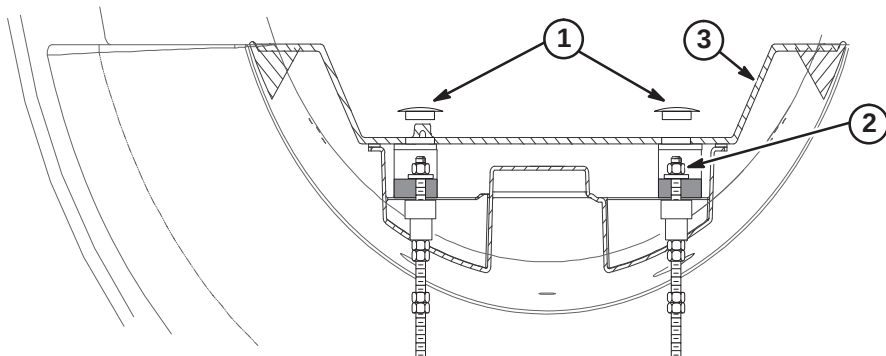


WARNING!

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.

4-2-3 Remove Front Split Bridge

1. At front of Bridge, remove hole plugs from each side of Bridge.
2. Remove M10 nut and washer from each stud.
3. Carefully lift and remove Front Split Bridge.

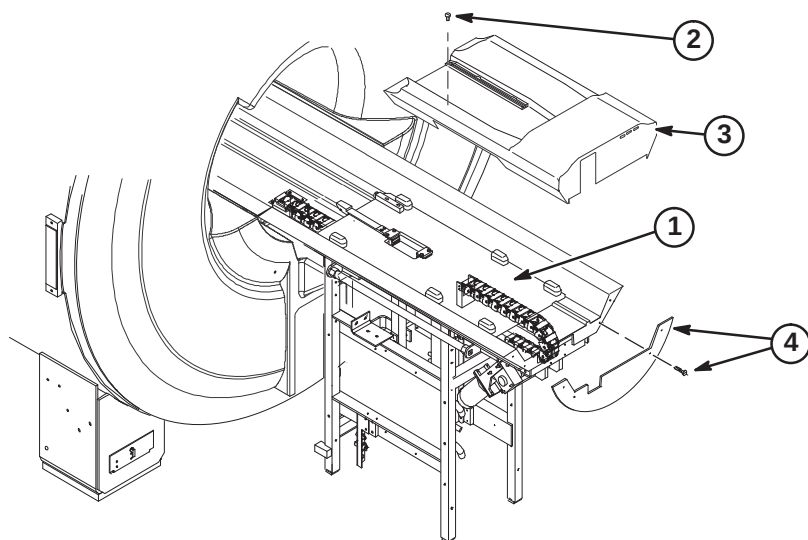


4-3 DE-INSTALLATION OF LONG BRIDGE

The procedures in this section provide instructions on removing the Long Bridge. For those systems that have a Split Bridge that needs to be removed, refer to Section 4-2.

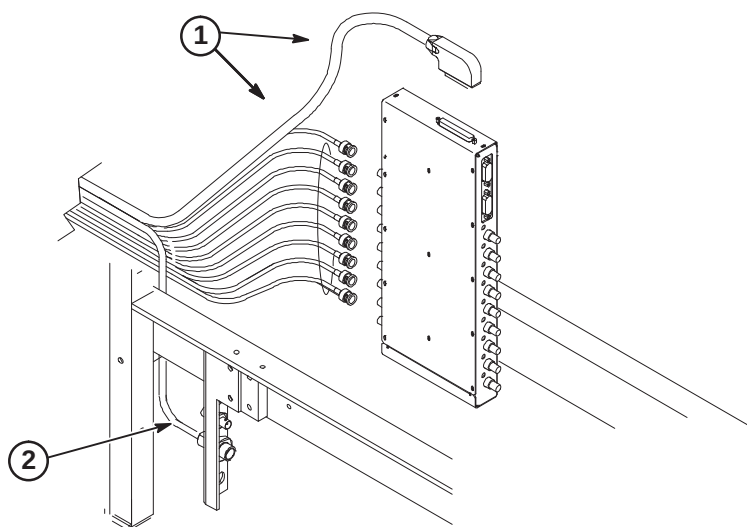
4-3-1 Remove LPCA and Bridge Covers

1. Make sure LPCA has been moved to rear of Bridge.
2. Remove the four screws that attach the cover to the base plate. Save screws for re-installation at end of upgrade.
3. Carefully remove cover from base plate assembly.
4. Unfasten four screws and remove Bridge Cover. Save for re-installation.



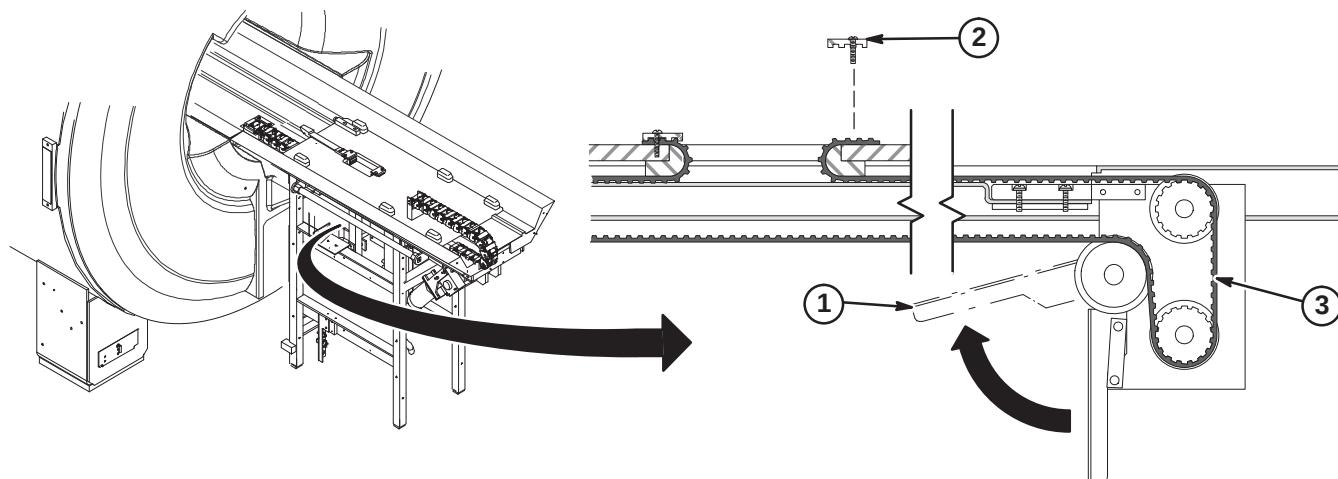
4-3-2 Disconnect LPCA cables in Rear Pedestal

1. Disconnect all cable routed from LPCA to 8-Channel Switch.
2. Disconnect Run 404 routed to Rear Pedestal I/F bracket.

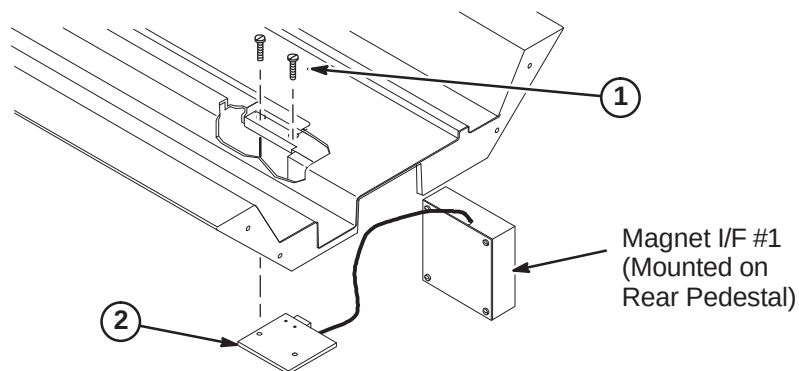


4-3-3 Disconnect Drive Belt from Rear Pedestal

1. Release Tension Arm in Rear Pedestal underneath Bridge.
2. Unfasten and save Clamp Block as shown below. The other Clamp Block holding the other end of the Drive Belt does not need to be removed.
3. De-install Drive Belt from around rear pulleys and Tension Arm pulley in Rear Pedestal. Secure Drive Belt to underside of Bridge to prevent damage when Bridge is removed.

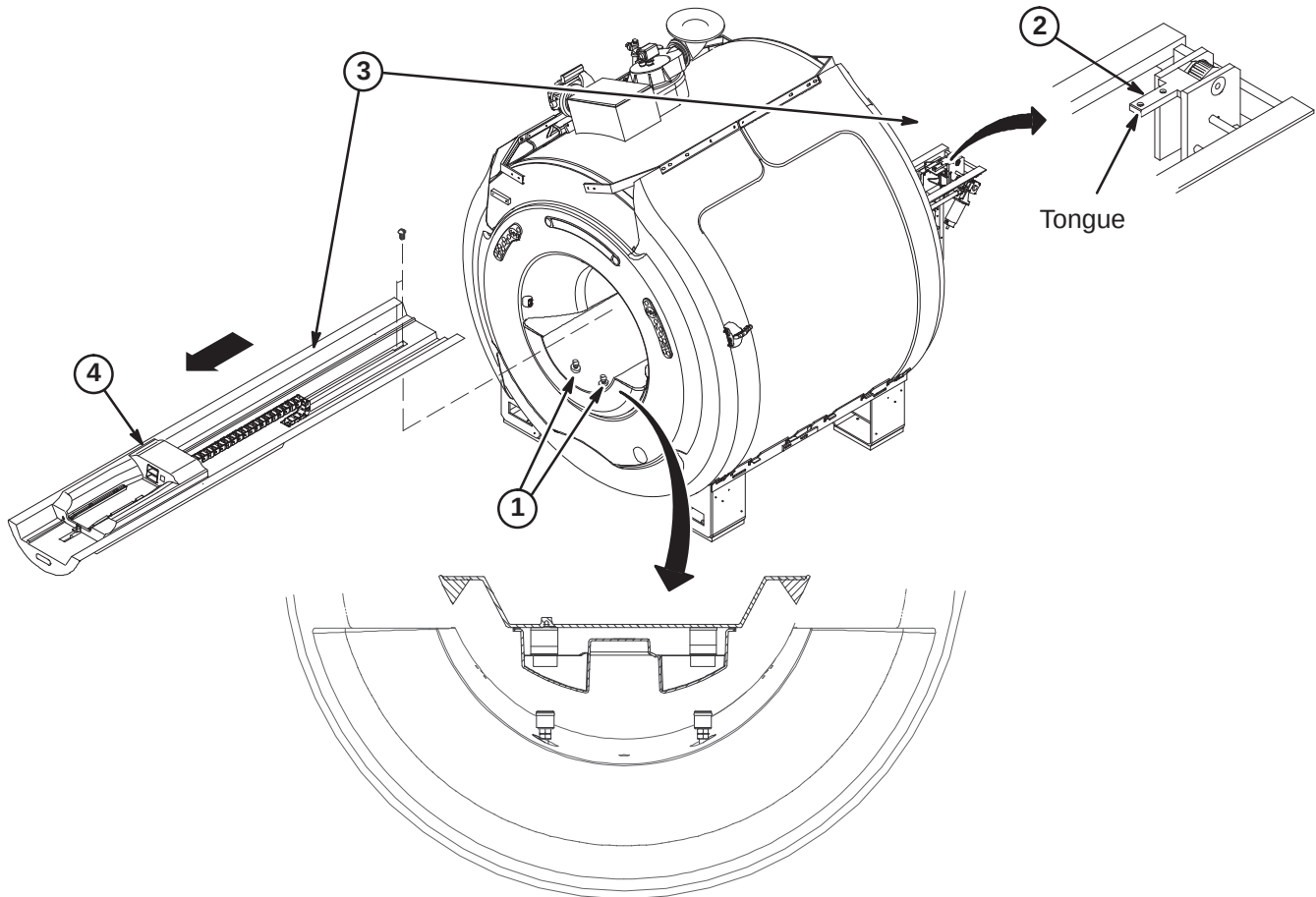
**4-3-4 Disconnect Bridge at Rear Pedestal**

1. Unfasten screws that hold Bridge, Tongue, and Rear Bridge Mount Assembly together.
2. Secure Rear Bridge Mount to prevent damage until Bridge is re-installed.



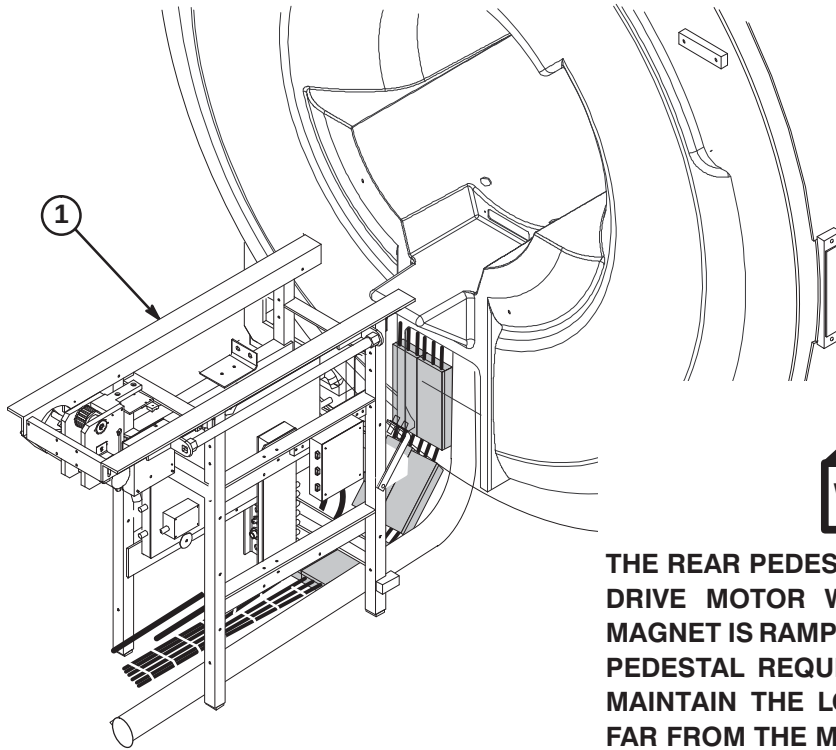
4-3-5 De-Install Long Bridge

1. Carefully lift front of Bridge from mounting studs. Support Caps at top of studs are glued to bottom of Bridge.
2. Lift front of Bridge to dis-engage tongue on from Bridge.
3. While one person holds the front end, the other person stands in back of the bore next to the bridge and guides the Bridge Asm through the bore towards the front.
4. Place Bridge with LPCA and attached cables in a secure location. They will be re-installed.



4-3-6 Move Rear Pedestal

1. Move the Rear Pedestal back, away from the magnet enclosure so that there will be enough room to unfasten and remove the Rear End Bell.

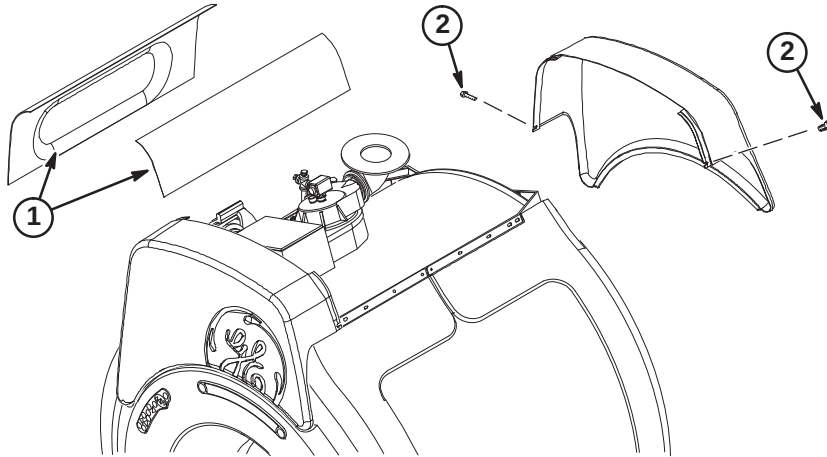
**WARNING!**

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.

4-4 DE-INSTALLATION OF MAGNET ENCLOSURE COVERS AND REAR END BELL

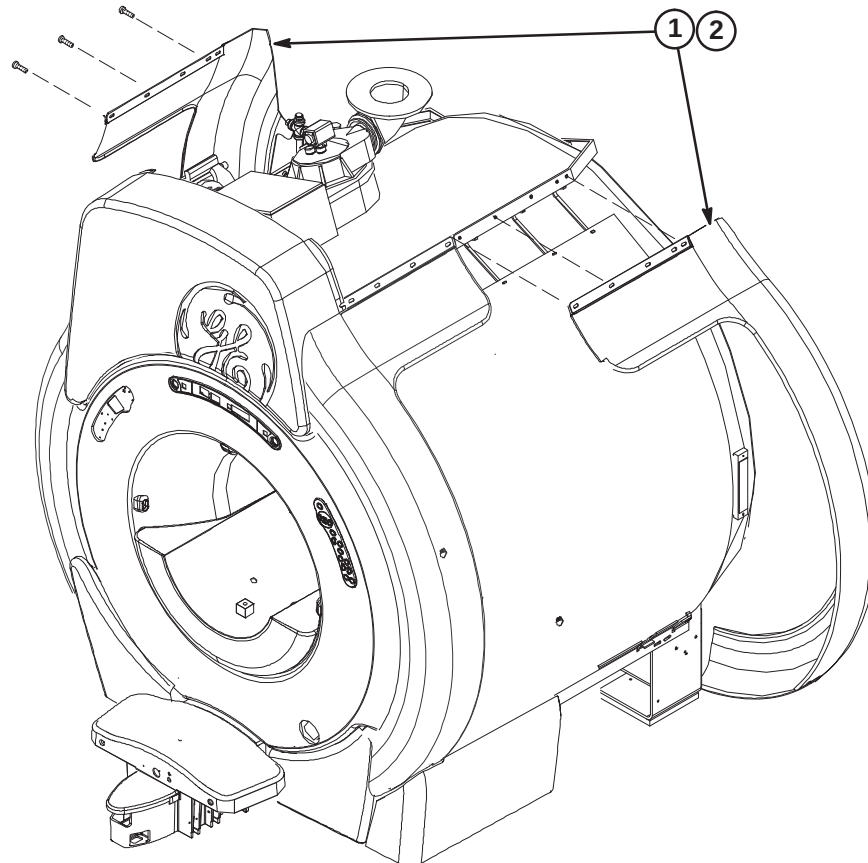
4-4-1 Remove Top Side Covers and Rear Cover

1. Carefully remove top left and right Side Covers.
2. De-install the Rear Top Cover by unfastening the M10 x 25mm screws holding it in place.



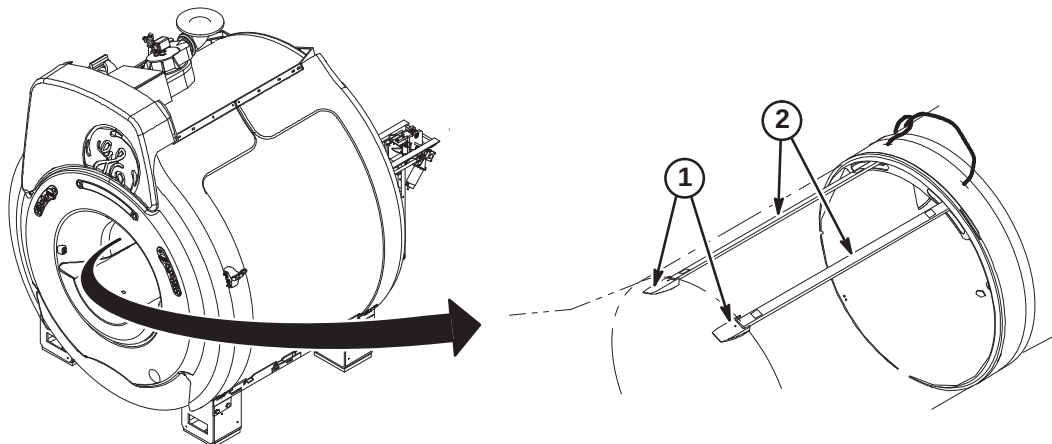
4-4-2 Remove Rear Arc Covers

1. Unfasten left and right Rear Arc Covers.
2. Carefully slide covers out from behind Rear End Bell and store in a safe place. Covers will be re-installed later.



4-4-3 Unfasten Bore Light Panels

1. Unfasten screws and remove Bore Light Panels at top of magnet bore.
2. Bore Light Panels are held in place with an adhesive. Carefully pull away the Bore Light Panels from the top of the bore. They will be reinstalled later.

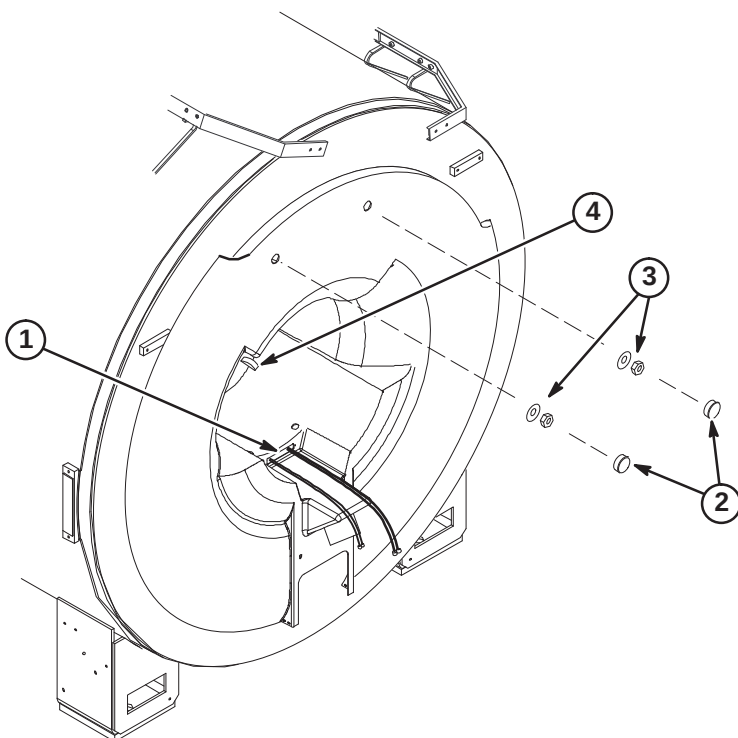


4-4-4 Unfasten Rear End Bell for Removal

1. Unfasten M4 x 10mm screws and remove Feed Thru Cover and Gasket over body coil cables.
2. Remove Access Hole Covers.
3. Unfasten and remove M10 Lock Nuts and Washers.

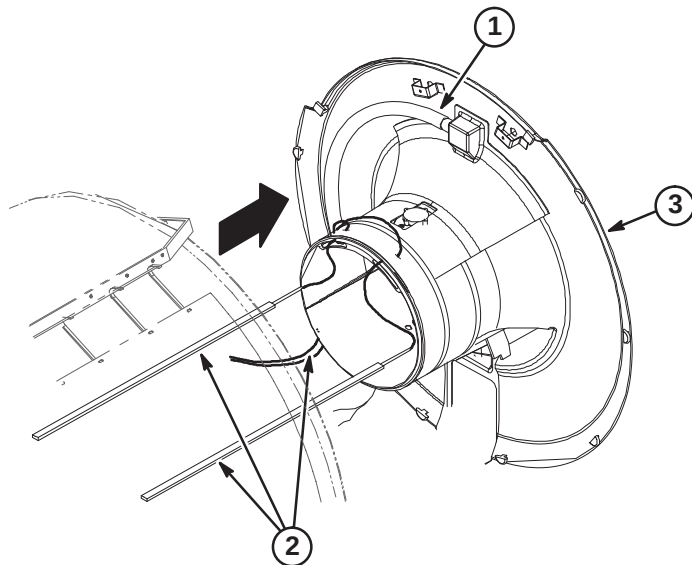
Note: Make sure there is enough room between magnet and Rear Pedestal for removal of End Bell and that there are at least two people holding End Bell for removal.

4. Unlatch two latches holding Rear End Bell to RF Coil.



4-4-5 Remove Rear End Bell

1. Move End Bell away from magnet and disconnect 2 inch air hose from End Bell.
2. Take care not to damage Bore Light panels or attaching Bore Lamp fiber optic cable.
3. Move Rear End Bell out of magnet room. This is required to perform the upgrade alterations.



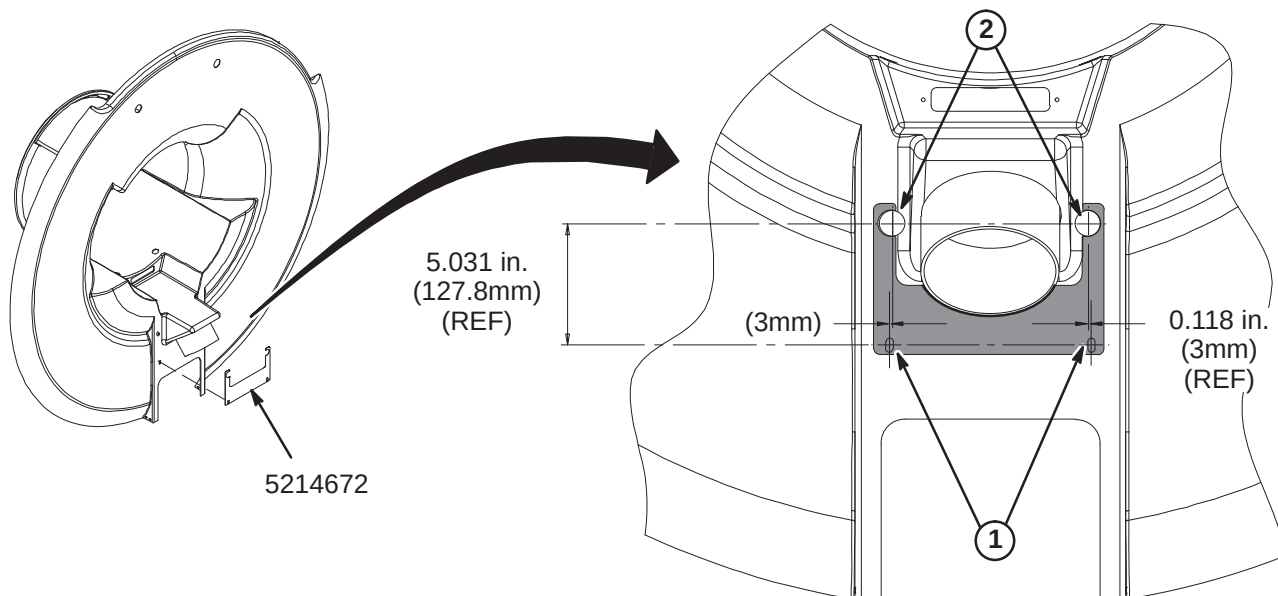
4-5 INSTALL STABILIZATION COMPONENTS

The components installed in this section are part of 5198961, the Pedestal Brace Kit assembly.

4-5-1 Create Clearance Holes in Rear End Bell

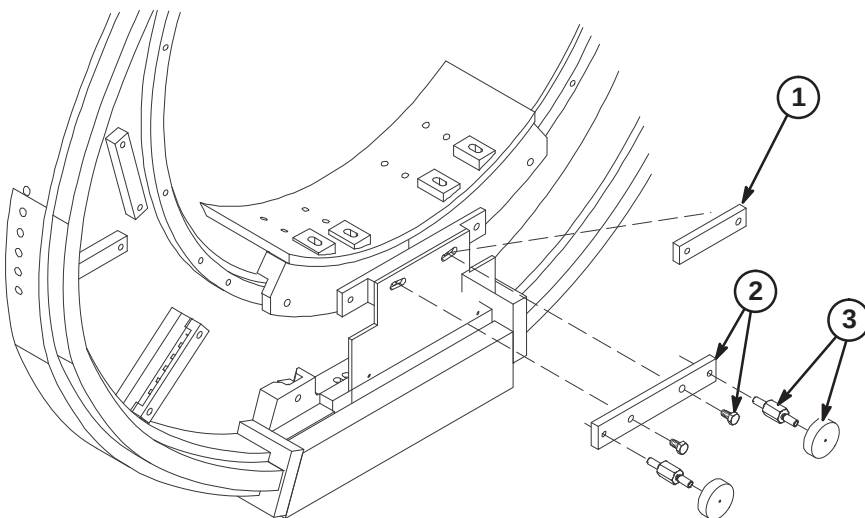
1. Attach End Bell Drill Template (5214672) to Rear End Bell using slots to align template.
2. Using template, cut 2 holes in End Bell 1.125 to 1.437 diameter (28,9 to 36,4) in diameter.

Note: Holes will cut into curved surfaces of End Bell.



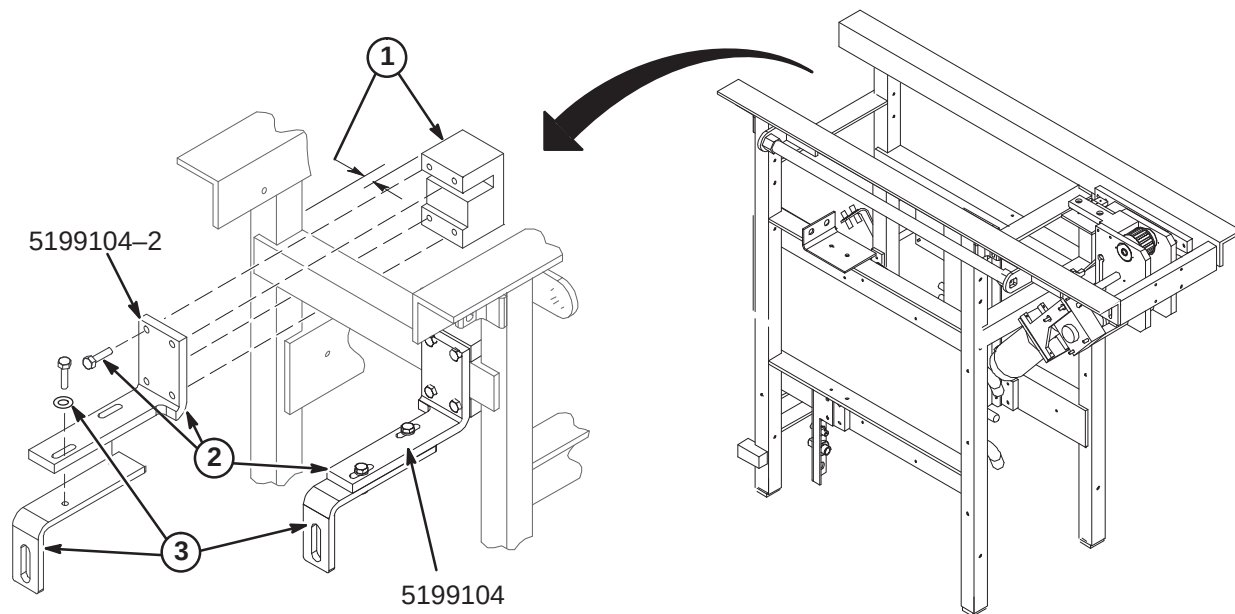
4-5-2 Install Mounting Block on Magnet

1. Unfasten hardware, remove and discard Clamp Block. Save fasteners.
2. Install new Washer Block (5112162) with fasteners removed in Step 1.
3. Install Hex Standoffs (5112148) and Foam Gaskets (2226752-2) as shown below. Adhesive side of foam must be positioned on side away from standoffs. Peel off protective paper.



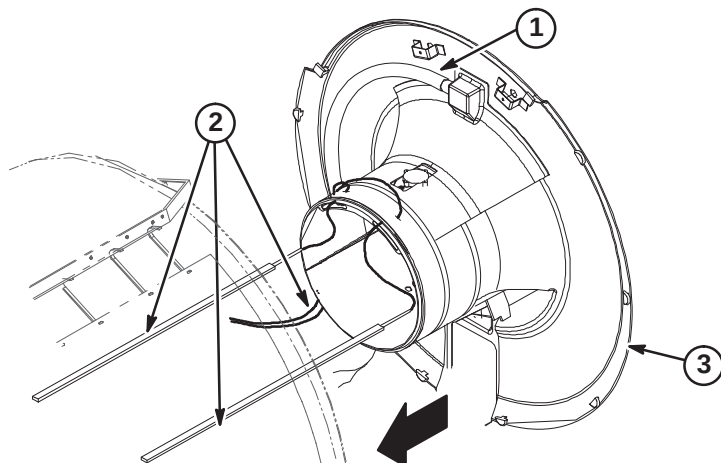
4-5-3 Install Rear Pedestal Clamp Block and Braces

1. Position the two Pedestal Clamp Blocks (5198645) approximately 7.5mm (0.30 in.) from the vertical rail on each side.
2. Secure each Pedestal Clamp Block to rail with Left Outer Pedestal Brace (5199104-2) and Right Outer Pedestal Brace (5199104) using four M6 x 20mm Hex Head Screws (2109866-12) on each side.
3. Loosely attach one Inner Pedestal Brace (5212104) to each Outer Pedestal Brace using two M6 x 20mm Hex Head Screws (2109866-12) and two plain Washers (2109878-3) on each brace.

**4-5-4 Install Rear End Bell**

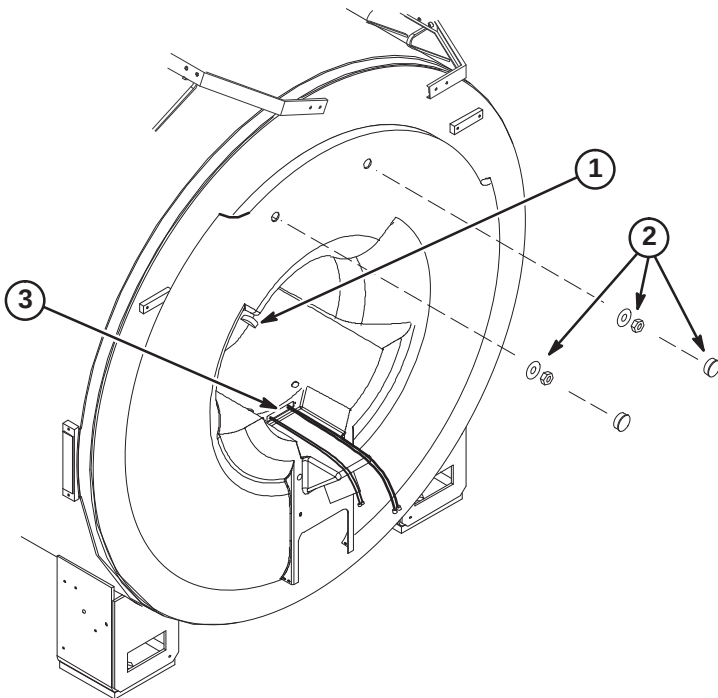
Note: Make sure there is enough room between magnet and Rear Pedestal for installation of End Bell and that there are at least two people holding End Bell.

1. Move End Bell into position at rear of magnet and connect 2 inch air hose from End Bell.
2. Position Bore Light panels in bore.
3. Carefully insert Rear End Bell into magnet bore.
 - Make sure the four RF Body Coil cables are routed through opening in Rear End Bell
 - Make sure Bore Light panels are move forward as End Bell is positioned.

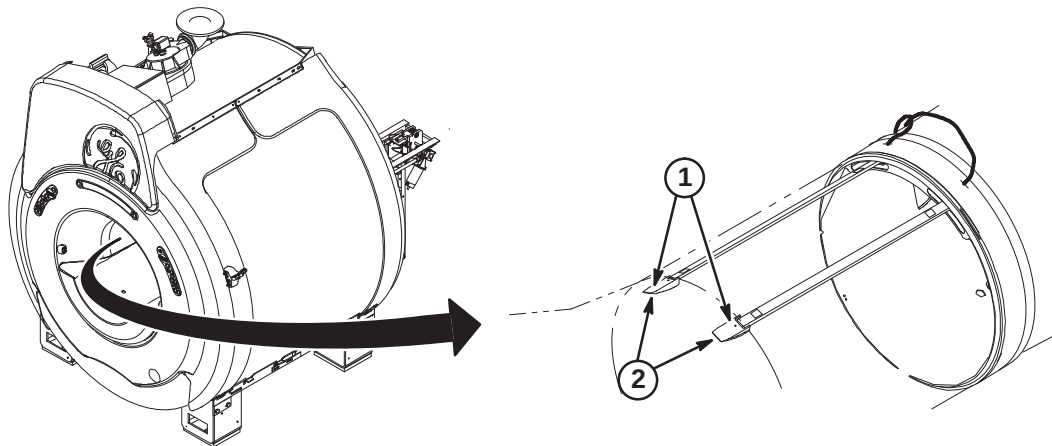


4-5-5 Fasten Rear End Bell to Magnet

1. Latch two latches between Rear End Bell to RF Coil.
2. Fasten M10 Lock Nuts and Washers and install Access Hole Covers.
3. Position Feed Thru Cover and Gasket over body coil cables and fasten with M4 x 10mm screws.

**4-5-6 Fasten Bore Light Panels**

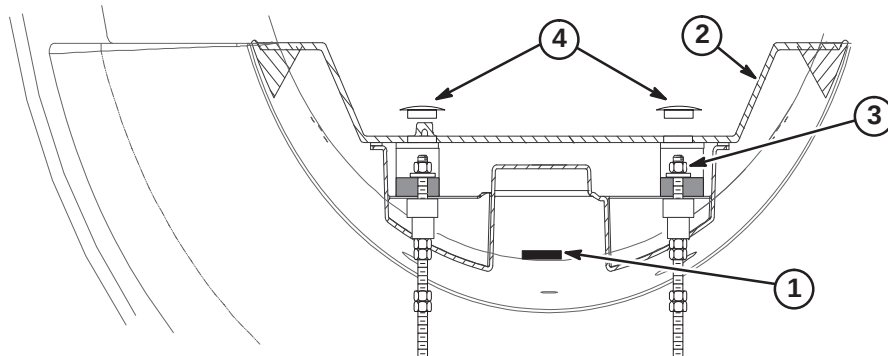
1. Insert Bore Light Panels into holder and press in place.
2. Fasten holders with previously loosened screws.



4-6 CONNECTION OF REAR PEDESTAL FOR SYSTEM WITH SPLIT BRIDGE

4-6-1 Install Front Split Bridge

1. Pull Drive Belt thru bore to front of magnet. Route Drive Belt around pulley of Front Split Bridge after Step 4.
2. Carefully position Front Split Bridge in bore of magnet.
3. Secure in place with previously removed M10 nuts and washers on each stud.
4. At front of Bridge, insert hole plugs on each side of Bridge.



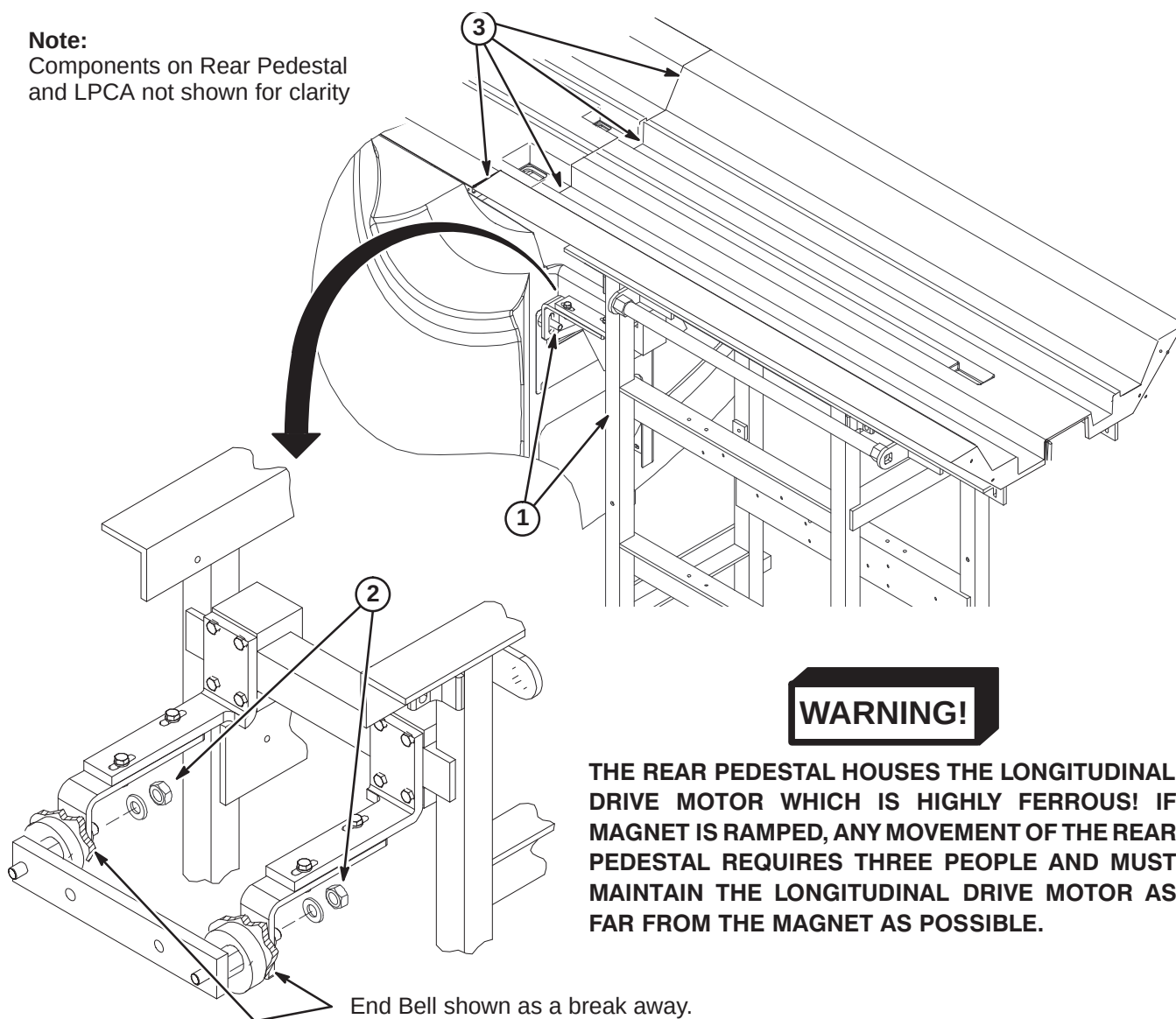
4-6-2 Connect Rear Pedestal to Standoffs**CAUTION**

The weight of the Rear Pedestal requires at least **THREE** people to move Rear Pedestal toward the rear of the magnet and position on standoffs.

1. Carefully move Rear Pedestal towards Rear End Bell and position installed pedestal braces on standoffs and the four studs on the Front Split Bridge with the holes in the Rear Split Bridge.
2. Connect, but do not tighten, left and right pedestal braces to standoffs using M10 Nuts (2109875-4) and M10 Washer (2109878-4) on each standoff.
Note: Fasteners will be tightened after Bridge is installed and aligned with Rear Pedestal.
3. From underneath Bridge, fasten the four hex nuts and washers that attach the Front and Rear Split Bridge.

Note:

Components on Rear Pedestal and LPCA not shown for clarity

**WARNING!**

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.

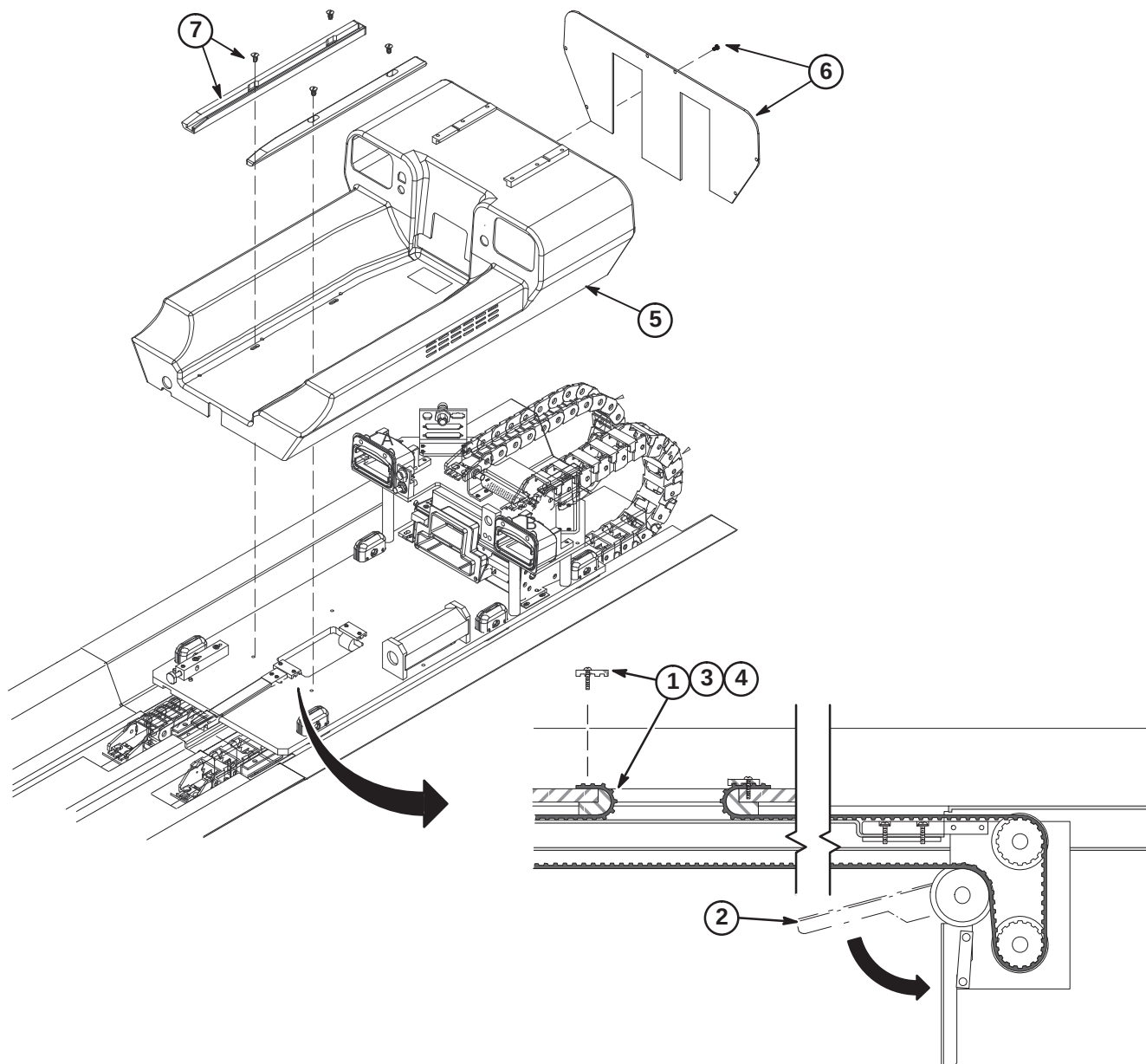
4-6-3 Re-Install LPCA Cover

Dual-track LPCA is shown below. Procedure would be the same for a system with a single-track LPCA.

1. Route end of Drive Belt from Front Split Bridge thru opening in LPCA Base Plate and attach with clamp.
2. Check Belt tension by closing Tension Arm.

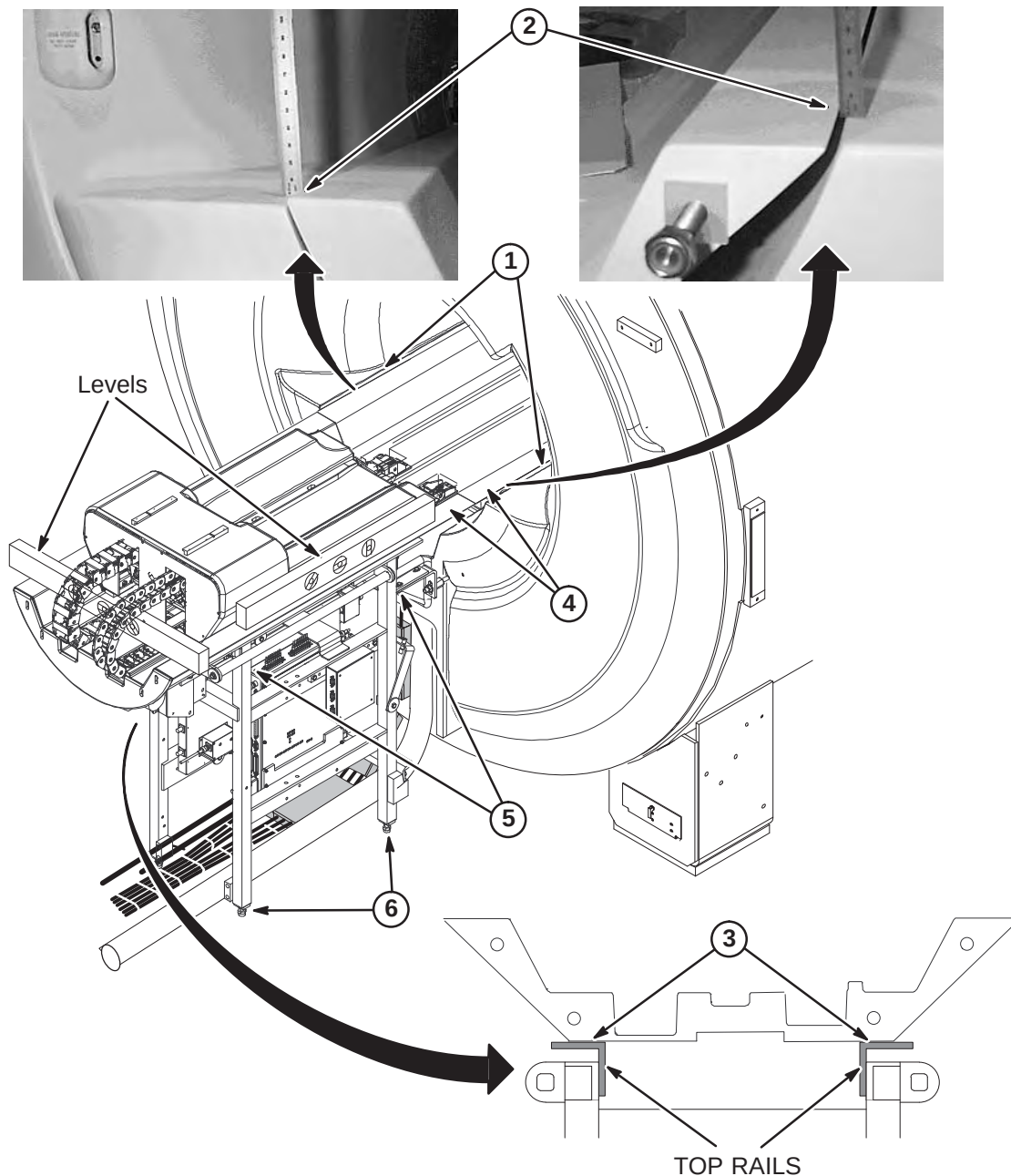
Note: Tension Arm should lock crisply into position without excessive force (which could cause later problems). If adjustment is necessary go to step 3. If not, go to step 4.

3. (IF NECESSARY): Loosen Clamp and adjust position of Belt one tooth at a time until desired tension is obtained.
4. Tighten Clamp and trim any excess Belt length.
5. Carefully place cover on base plate assembly.
6. Attach Rear Cover using 6 screws previously removed.
7. Attach the cover to the base plate using four screws previously removed.



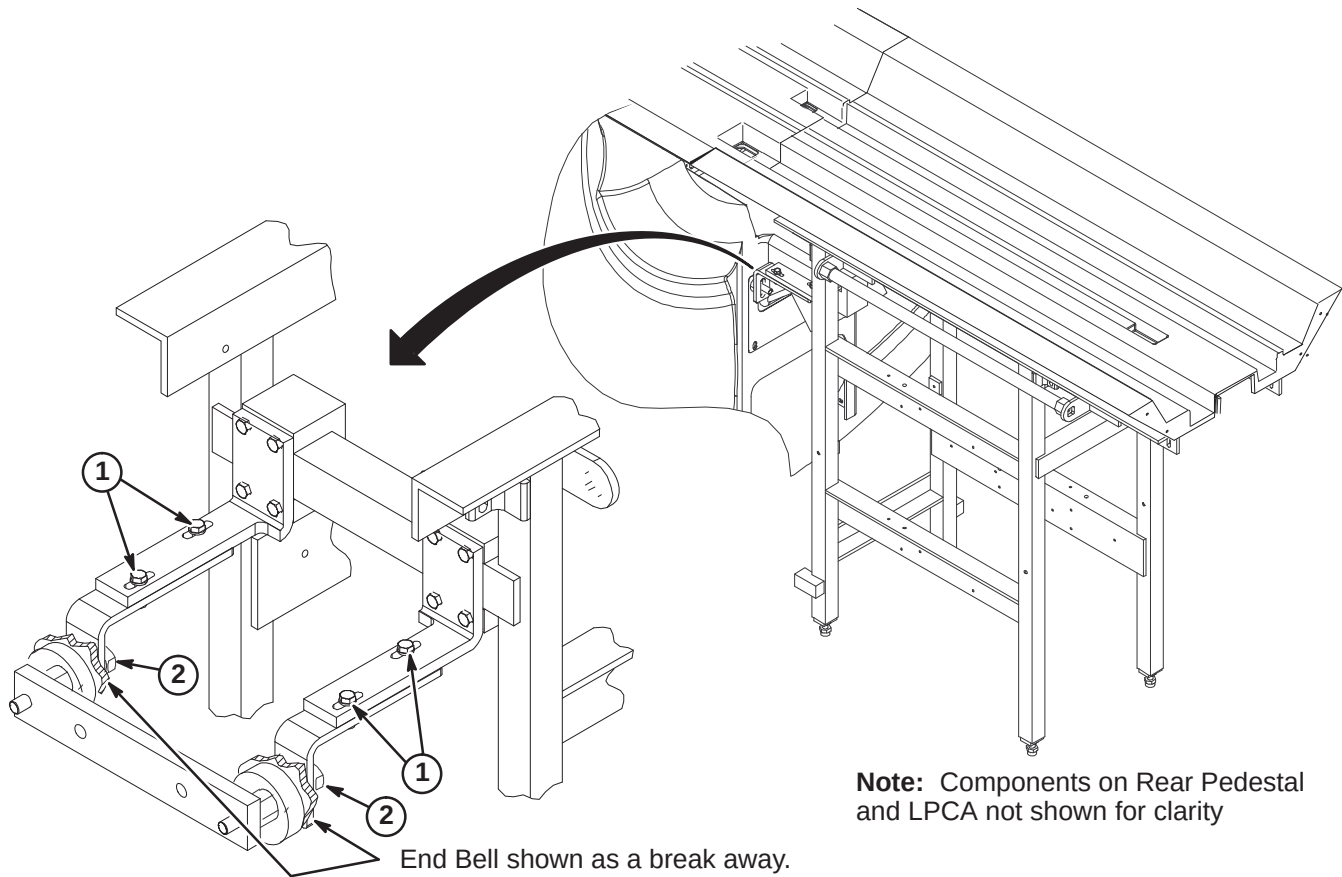
4-6-4 Level and Align Bridge

1. Align Rear Pedestal and Split Bridge to magnet enclosure. Spacing between sides of Bridge and bore of magnet should be equal.
2. Level Bridge and make sure that Bridge surface is no less than 2mm and no more than 4mm above the surface of front and rear End Bells, as indicated in photos.
3. Rear Pedestal must be adjusted to maintain full contact of Rear Pedestal top rails with the bottom of the Bridge along entire length of the top rails. Use vertical adjustment fasteners at top of each leg to provide contact.
4. Make sure that surfaces between front and rear Bridge surfaces are even. This will provide smooth travel for the Carriage wheels
5. Tighten vertical adjustment fasteners.
6. Adjust leg screws on both sides as needed to level Rear Pedestal.



4-6-5 Fasten the Inner and Outer Rear Pedestal Braces

1. Tighten the four screws that connect the Inner and Outer Pedestal Braces.
2. Tighten nuts on each standoff .



4-7 CONNECTION OF REAR PEDESTAL FOR SYSTEM WITH LONG BRIDGE

4-7-1 Connect Rear Pedestal to Standoffs

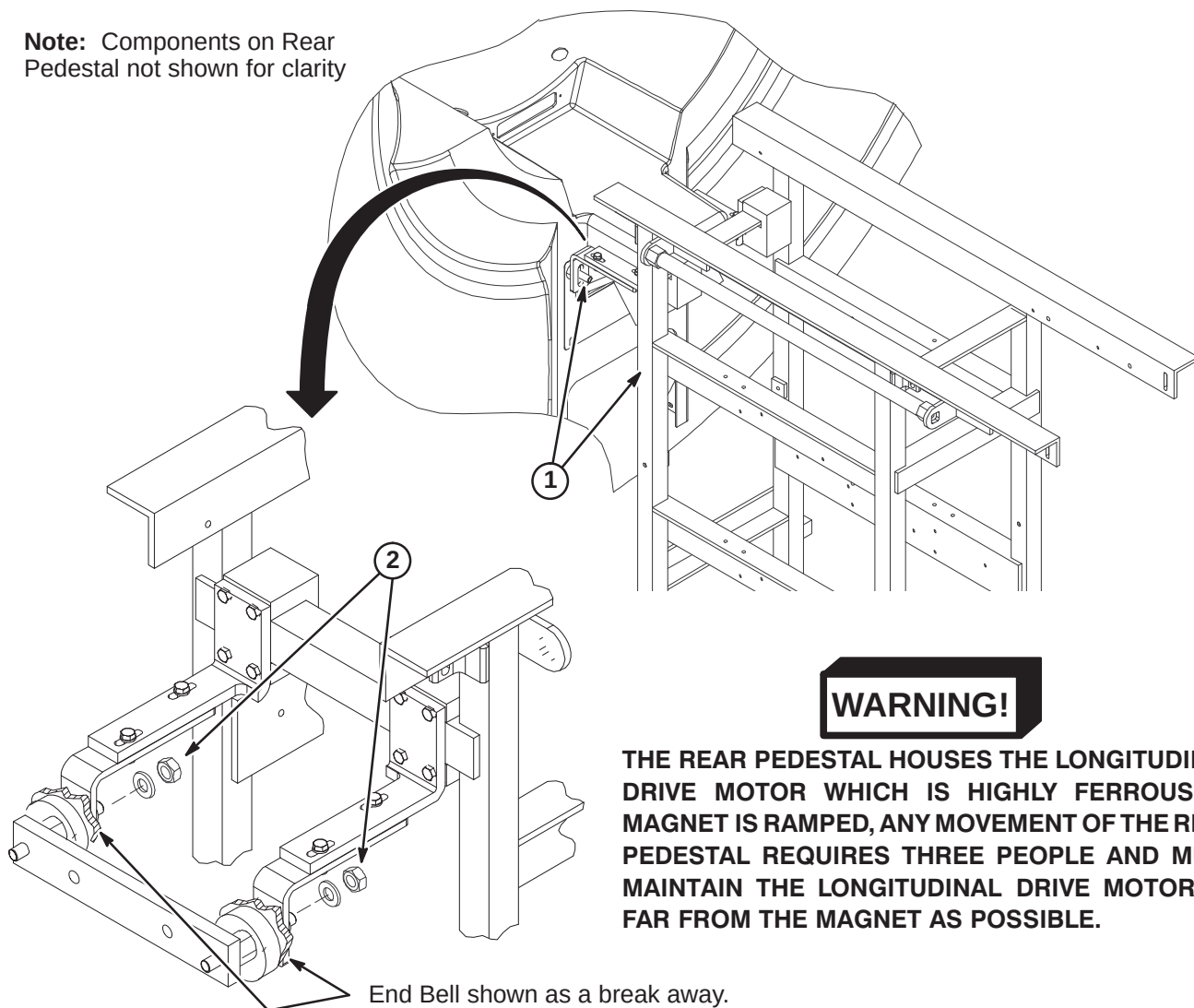


The weight of the Rear Pedestal requires at least **THREE** people to move Rear Pedestal toward the rear of the magnet and position on standoffs.

1. Carefully move Rear Pedestal towards Rear End Bell and position installed pedestal braces on standoffs and the four studs on the Front Split Bridge with the holes in the Rear Split Bridge.
2. Connect, but do not tighten, left and right pedestal braces to standoffs using M10 Nuts (2109875-4) and M10 Washer (2109878-4) on each standoff.

Note: Fasteners will be tightened after Bridge is installed and aligned with Rear Pedestal.

Note: Components on Rear Pedestal not shown for clarity



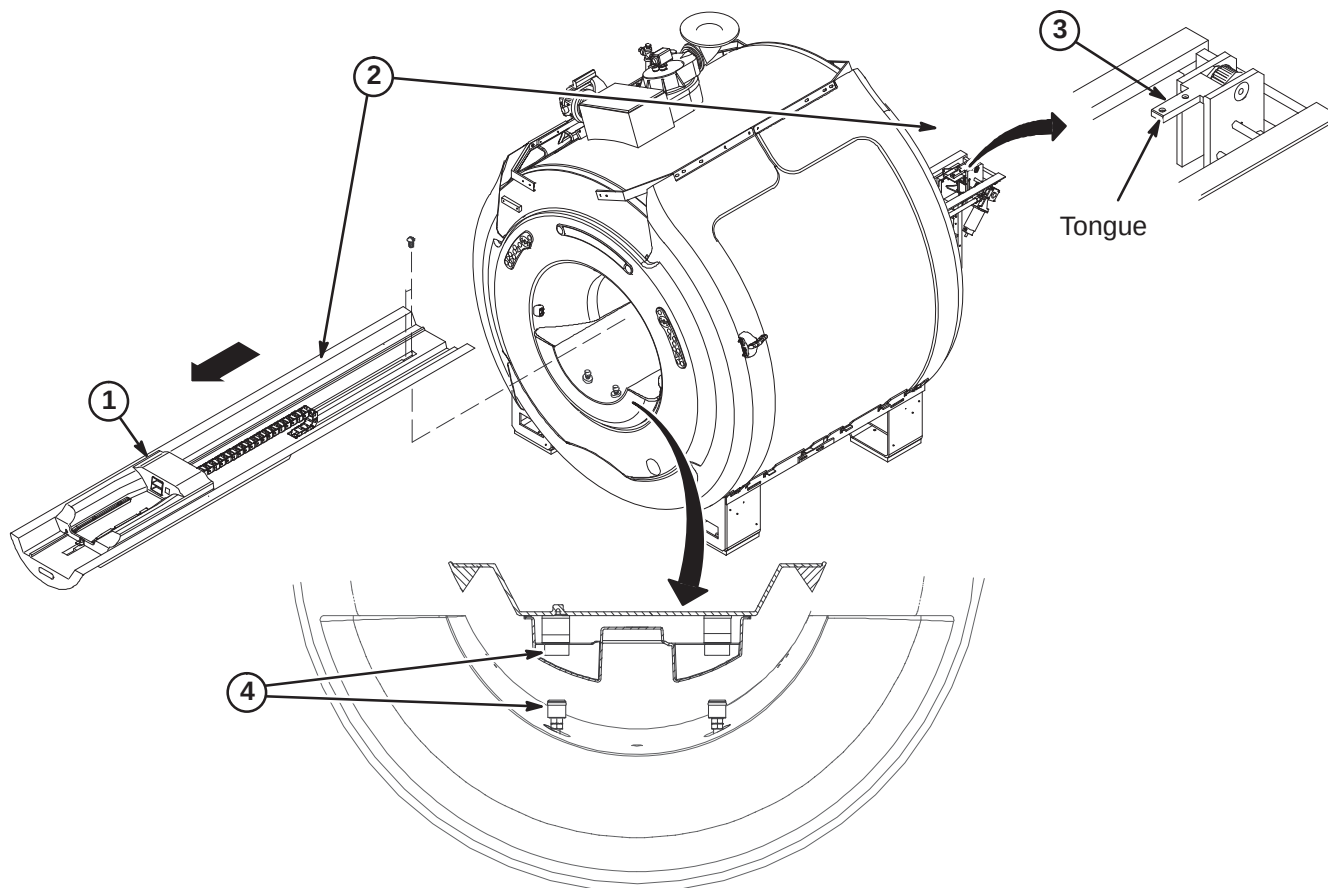
WARNING!

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.

4-7-2 Install Long Bridge

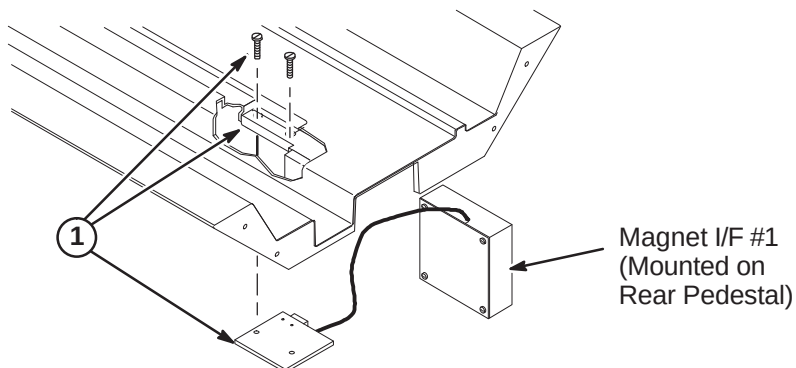
1. Position rear end of Bridge Asm at front of magnet enclosure.
2. While one person holds the front end, the other person stands in front of the bore next to the bridge and guides the Bridge Asm through the bore to the other end.
3. Lift front end of bridge to engage tongue on long drive assembly into opening at end of bridge.
4. Lower the front of Bridge onto the Bridge Support mounting studs. The Support Cap Cups glued to bottom of Bridge should align and slide onto the Support Caps at top of studs.

Note: If Support Cap Cups and Support Caps do not align, reposition Rear Pedestal as needed.



4-7-3 Connect Bridge at Rear Pedestal

1. Attach together Bridge, Tongue and Rear Bridge Mount Asm using two (2) previously removed 10-32 x 1-3/8 Screws (46-220184P47)

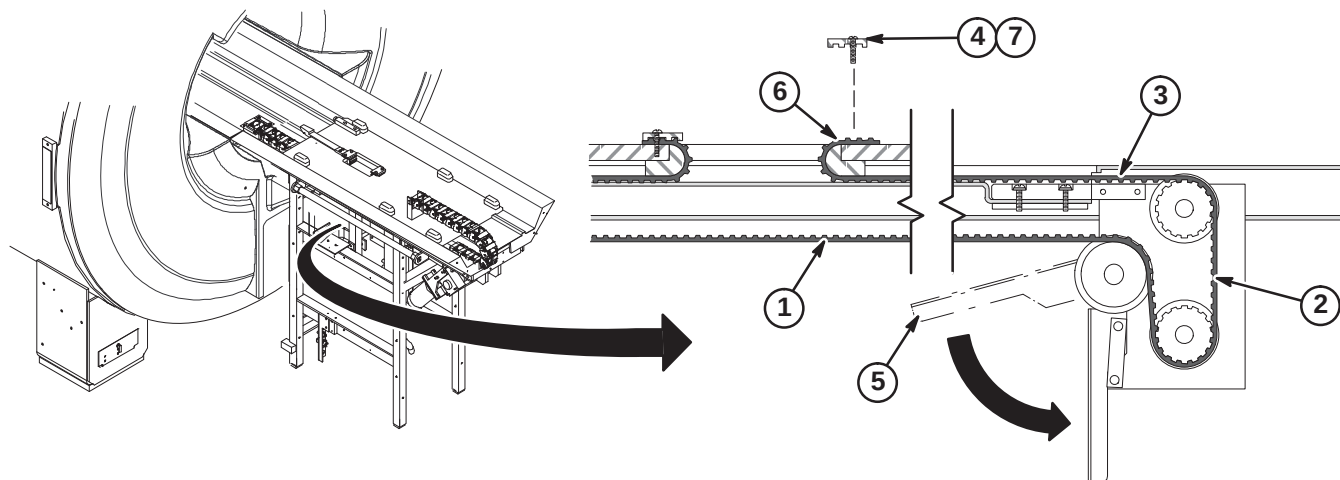


4-7-4 Connect Drive Belt in Rear Pedestal

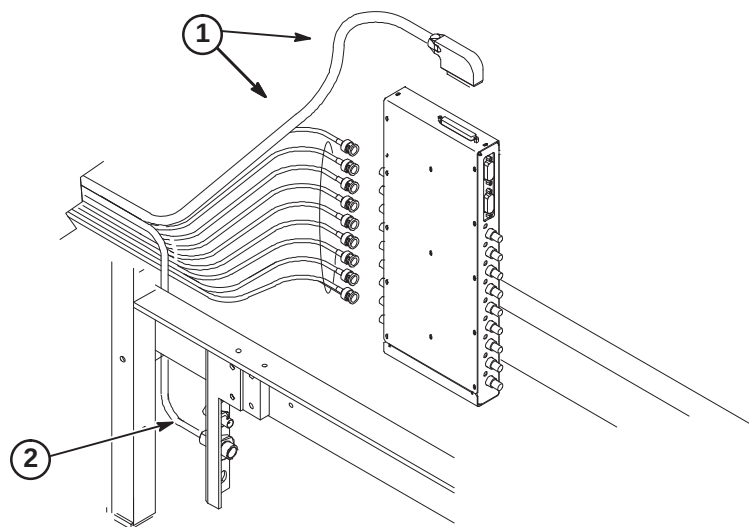
1. Route end of drive belt from front of Bridge towards Tension Arm Pulley.
2. Feed back end of Belt underneath Bridge, around Tension Arm Pulley and rear pulleys into Bridge.
3. Route Belt under and through opening to top of Carriage.
4. Position Belt under Clamp Plates and re-attach with previously removed screws.
5. Check Belt tension by closing Tension Arm. (See Note)

Note: Tension Arm should lock crisply into position without excessive force (which could cause later problems). If adjustment is necessary go to step 6. If not, go to step 7.

6. (IF NECESSARY): Loosen Clamp and adjust position of Belt one tooth at a time until desired tension is obtained.
7. Tighten Clamp and trim any excess Belt length.

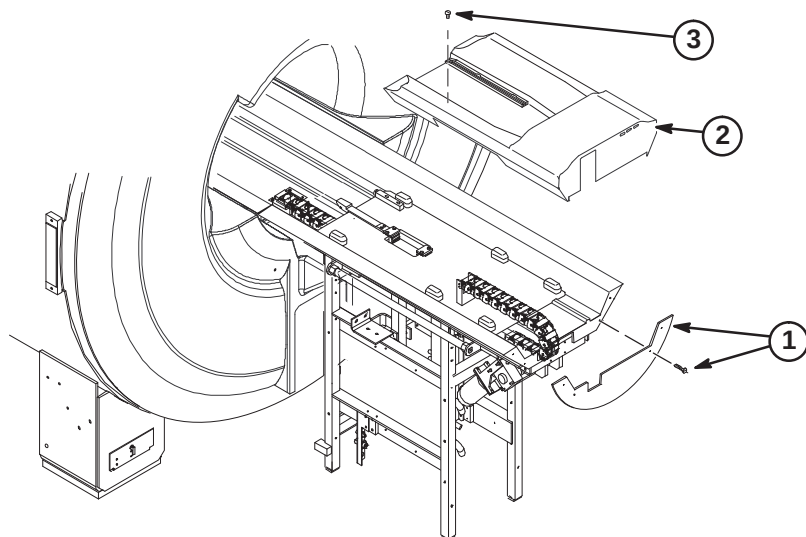
**4-7-5 Connect LPCA cables in Rear Pedestal**

1. Connect all cables routed from LPCA to 8-Channel Switch.
2. Connect Run 404 routed to Rear Pedestal I/F bracket.



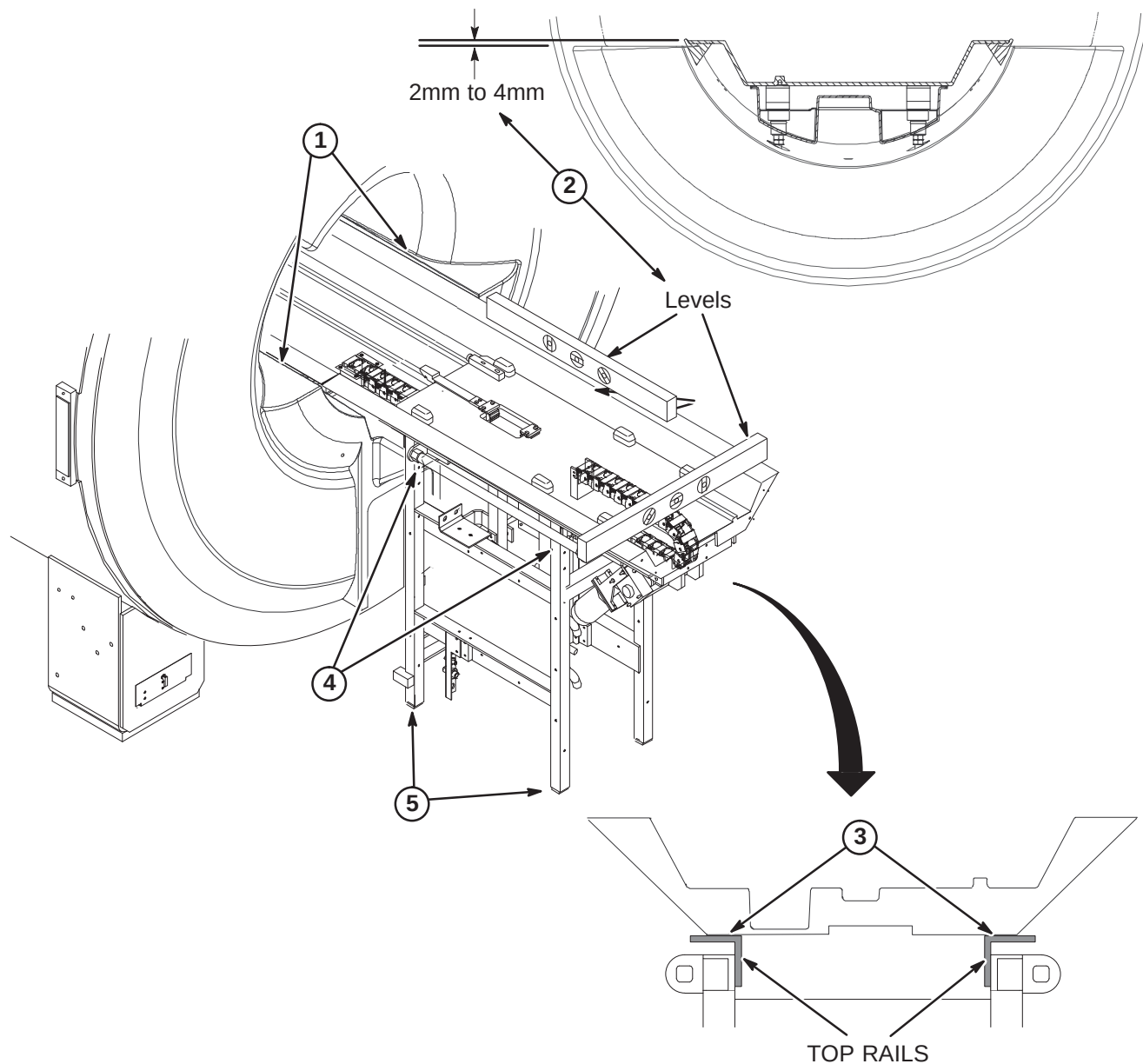
4-7-6 Install LPCA and Bridge Covers

1. Replace previously removed Bridge Cover and fasten with (4) screws to end of Bridge.
2. Position cover on base plate assembly.
3. Attach the cover to the base plate using four screws previously removed.



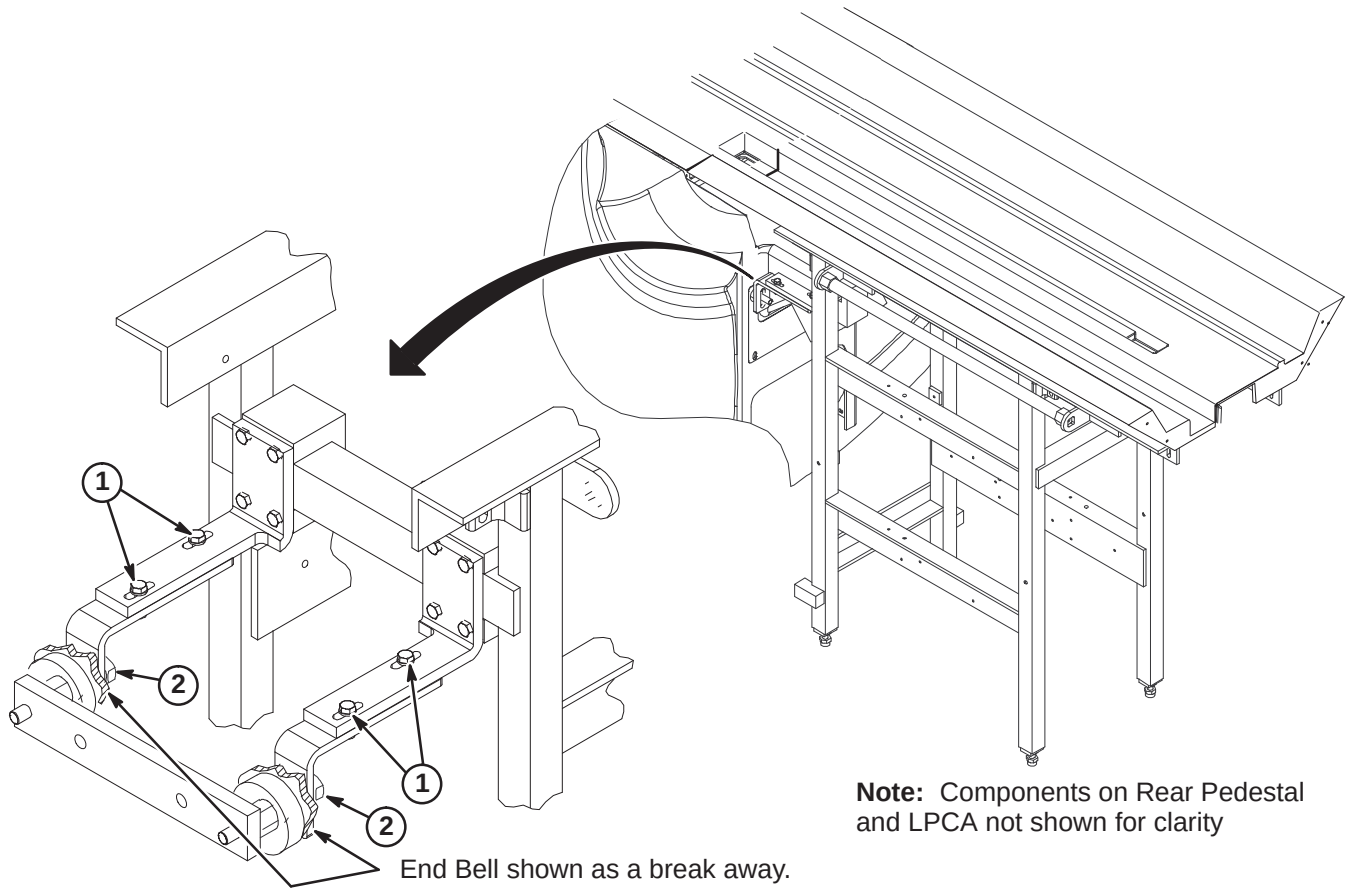
4-7-7 Level and Align Long Bridge

1. Align Rear Pedestal and Long Bridge to magnet enclosure. Spacing between sides of Bridge and bore of magnet should be equal.
2. Level Bridge and make sure that Bridge surface is no less than 2mm and no more than 4mm above the surface of front and rear End Bells, as indicated in photos.
3. Rear Pedestal must be adjusted to maintain full contact of Rear Pedestal top rails with the bottom of the Bridge along entire length of the Top Rails. Use vertical adjustment fasteners at top of each leg to provide contact.
4. Tighten vertical adjustment fasteners.
5. Adjust leg screws on both sides as needed to level Rear Pedestal.



4-7-8 Fasten the Inner and Outer Rear Pedestal Braces

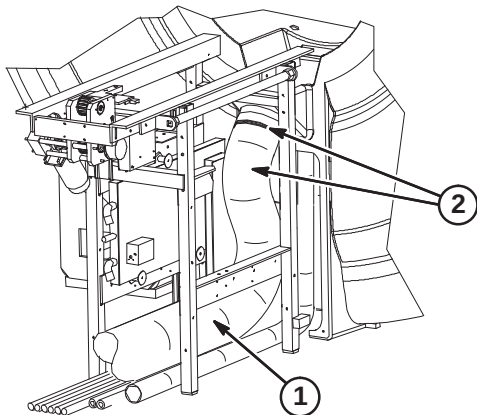
1. Tighten the four screws that connect the Inner and Outer Pedestal Braces.
2. Tighten nuts on each standoff .



4-8 COMPLETE INSTALLATION OF CABLES AND COVERS

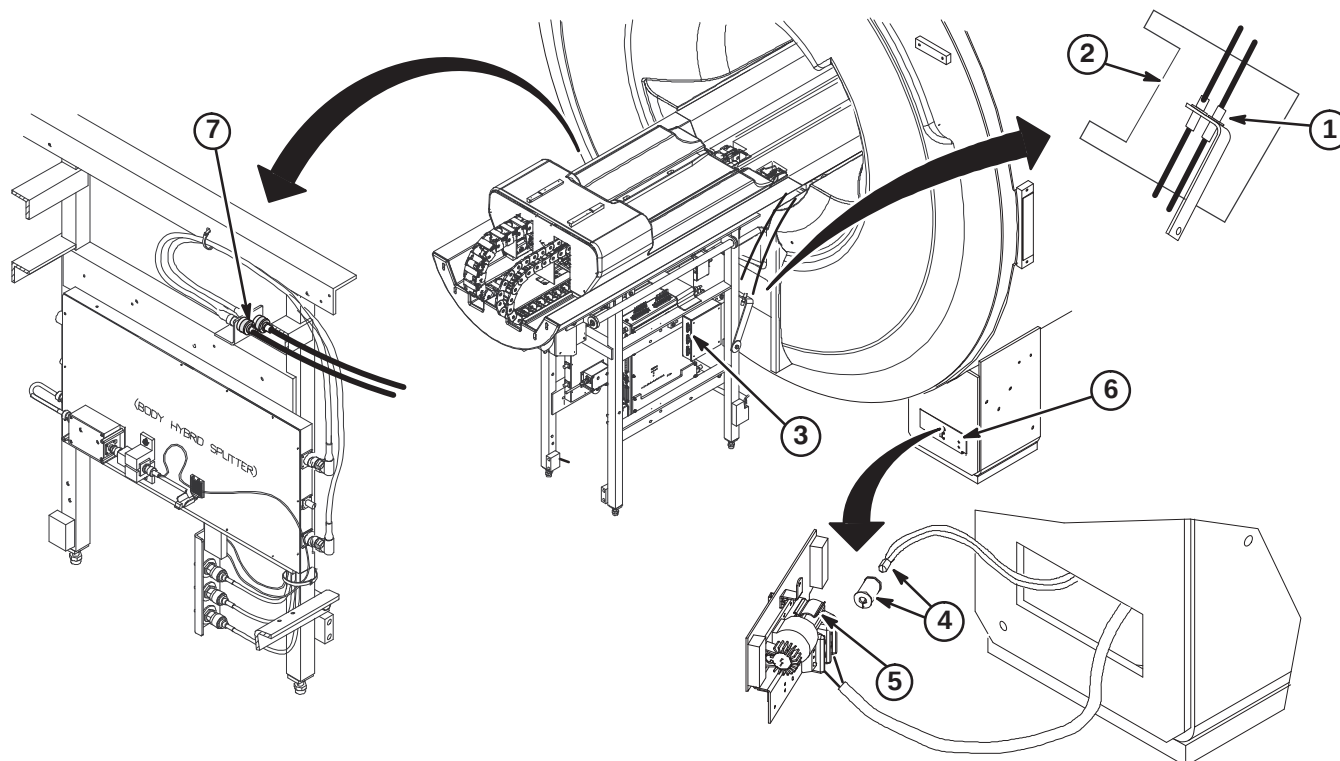
4-8-1 Re-Connect Air Hose

1. Route 6 inch hose back thru Rear Pedestal to Rear End Bell.
2. Slide 6 inch hose around Rear End Bell housing and secure with previously removed 6 inch Clamp.



4-8-2 Re-Connect Cables to Rear Pedestal Area

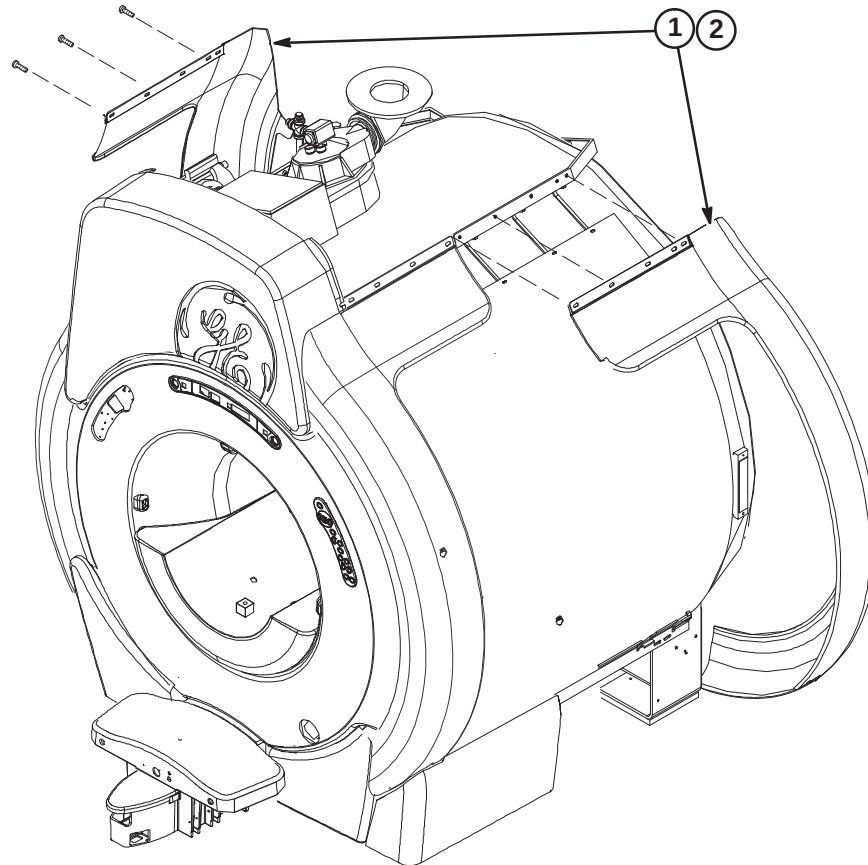
1. Connect RF body coil Bias cables to J5 and J6 on bracket.
2. Reuse Connector Wrap and cover Bias cable connections.
3. Re-connect cables to J8 and J9 on Patient Communication Box.
4. Insert Bore Light fiber optic into bushing and fasten with set screws.
5. Insert Bushing into clip on Bore Light Assembly. Bushing flanges must be inside clip.
6. Install Bore Light Assembly into rear leg of magnet.
7. Re-connect RF body coil I and Q cables to bracket.



4-9 INSTALL MAGNET ENCLOSURE AND REAR PEDESTAL COVERS

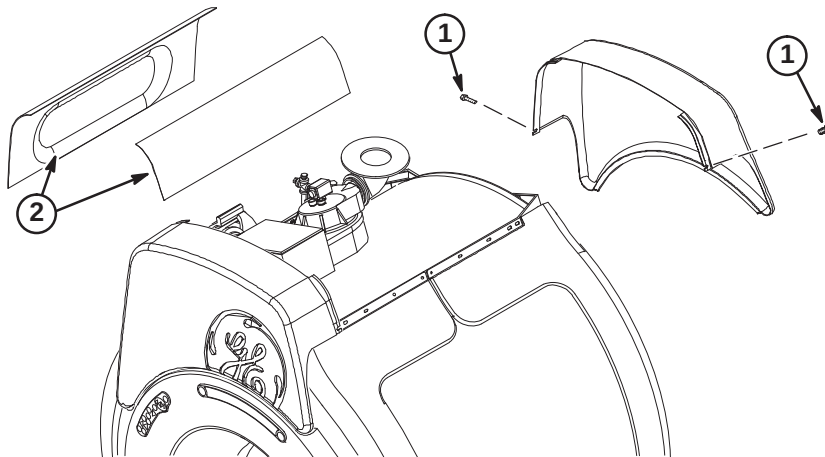
4-9-1 Install Rear Arc Covers

1. Carefully slide covers into position behind Rear End Bell.
2. Fasten left and right Rear Arc Covers using previously removed fasteners.



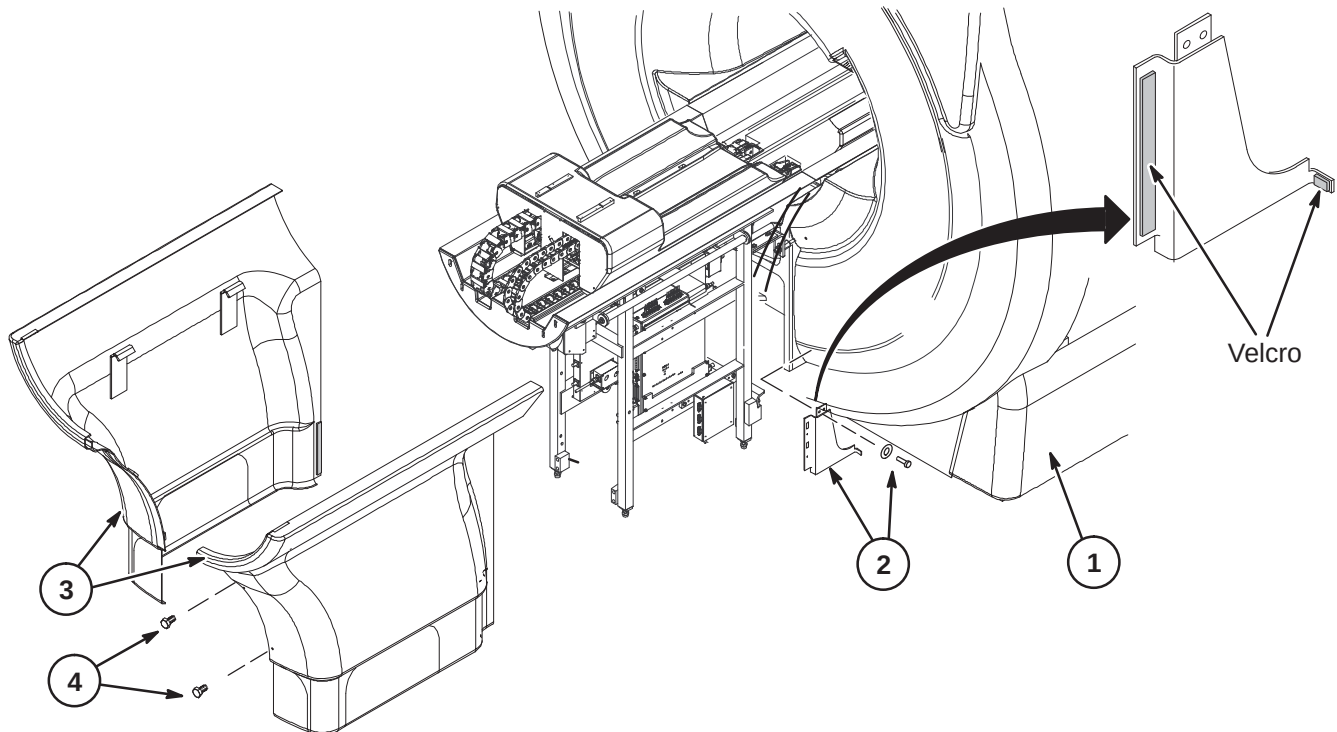
4-9-2 Install Top Side Covers and Rear Cover

1. Install the Rear Top Cover using the M10 x 25mm screws previously removed.
2. Carefully install top left and right Side Covers.



4-9-3 Install Rear Pedestal and Trim Covers

1. Install left and right rear lower trim covers.
2. Install Rear End Bell covers on each side of Rear Pedestal using the two M4 x 8mm screws and washers previously removed. Covers are attached to base trim with Velcro.
3. install left and right Rear Pedestal Covers on horizontal support bars. Covers are attached to base trim covers with Velcro.
4. Fasten left and right covers using two M6 x 10mm Pan Head screws as shown below.



SECTION 5 – COMPLETION

5-1 MECHANICAL INSTALLATION COMPLETION

The following steps need to be reviewed and completed by GE Field Engineer, especially if the system was installed by non-GE personnel. Check off boxes at the front of each item below when complete.

- ☐ Check mechanical flowchart to make sure all steps are complete (Section 2)
- ☐ Resolve shipment shortages
- ☐ Resolve omissions by mechanical contractors
- ☐ Make sure all rating plates are installed
- ☐ Recheck all wiring connections using cable map (Section 4)
- ☐ Correct for wiring errors if necessary
- ☐ Complete Sections 5-2, 5-3, and 5-4.

5-2 RETURNING SHIPPING EQUIPMENT

The shipping material must be returned **as soon as possible**. Prompt return is required so that production shipments can continue with these labor saving dollies. These can be included with the obsolete equipment and materials being returned in Section 5-3. Recycling Operation will route them to appropriate location.

5-3 MATERIAL DISPOSITION GUIDANCE

Refer to Section 1-6. The equipment or material being disposed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

5-4 FINAL COORDINATION

1. Replace cabinet covers that have not been previously replaced.
2. Dock Patient Transport in position at front of bore.
3. Record and enter applicable data into applicable site configuration files and records.
4. Complete Product Locator information. See Section 1-7.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

5. Leave the site's set of Manuals and CD-ROM at site. Organize and set up reference cabinet for the Manuals.
6. Locate any new Material Safety Data Sheets (MSDS). They must be retained on site. Customer is to be informed that material with MSDS information was brought into site and customer should know/decide where MSDS be retained.
7. Verify that coordination of application support for instruction of site users on Software Release 14.x has been made before returning system to the users. Turn site over to applications who will instruct users.

5-5 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Refer to the CD-ROM, Installation Calibration Flow, and perform the following:
 - PDU Voltage Checks Compact > PDU E-Stop Check
 - Startup Checks Compact
4. On the PDU:
 - A. Move the mMain Input Circuit Breaker operating handle to OFF position (if in tripped state).
 - B. Move to ON (vertical) position.
 - C. Press the EMO Reset button.
5. Except for the HFD Gradient power modules, switch all cabinet circuit breakers to ON. The HFD Gradient power module circuit breakers should remain OFF for one hour after HVAC power up to allow room humidity to stabilize.
6. **Make sure the UPS circuit breaker is switched to the ON position.** This is required to feed power to the GOC.